Physical Filters, Colored Light, Inputs/Outputs, Communication & Troubleshooting

Lecture 6

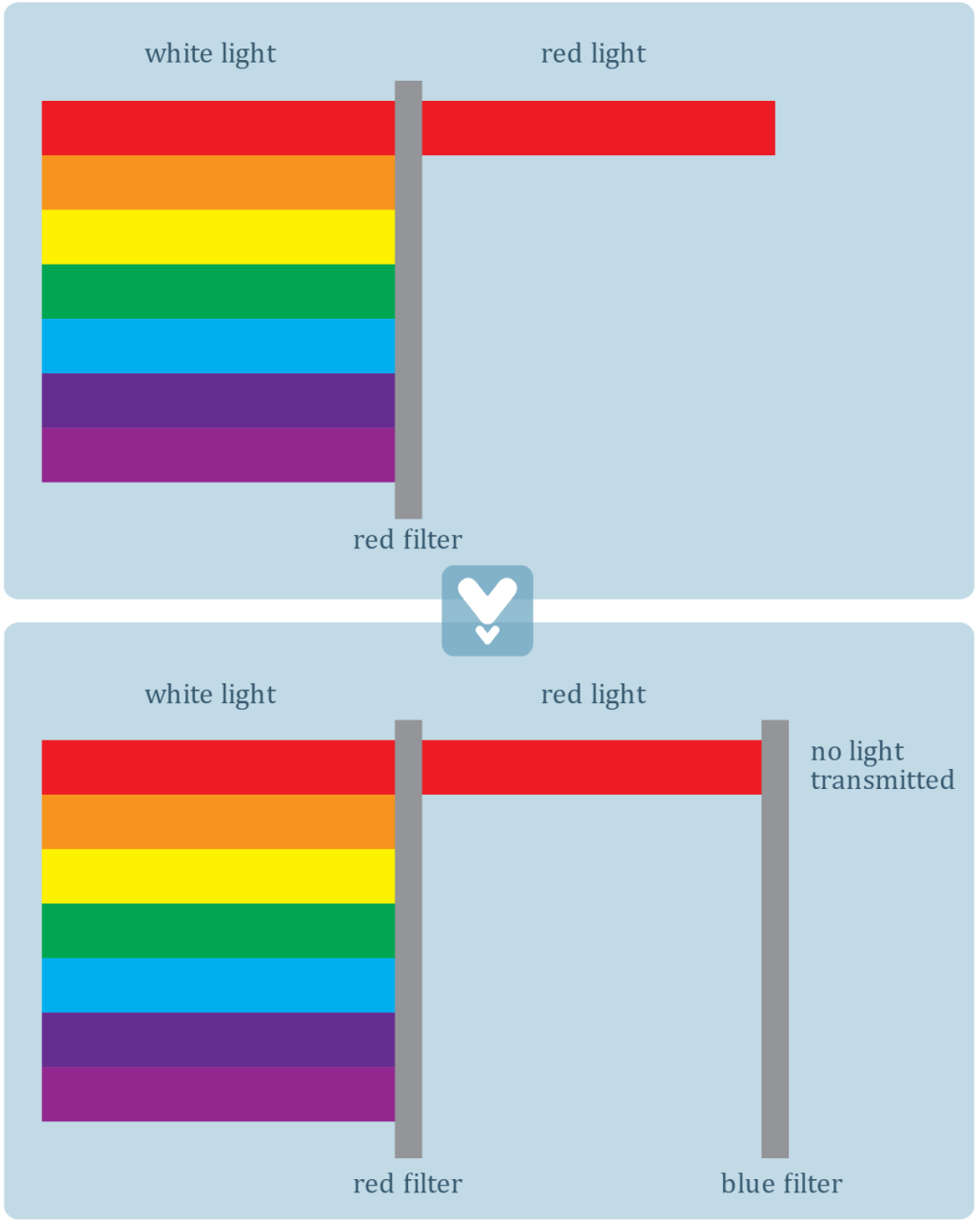
Physical Filters & Colored Light

Digital vs. Physical Filters

- Digital Filters: process **pixel values after** the image is captured.
  - Binarize
  - Clip
  - Invert
  - Optical Density Filters, etc.
- Physical Filters: control **light before** it hits the camera sensor.
  - Color Filters
  - Polarizing Filters

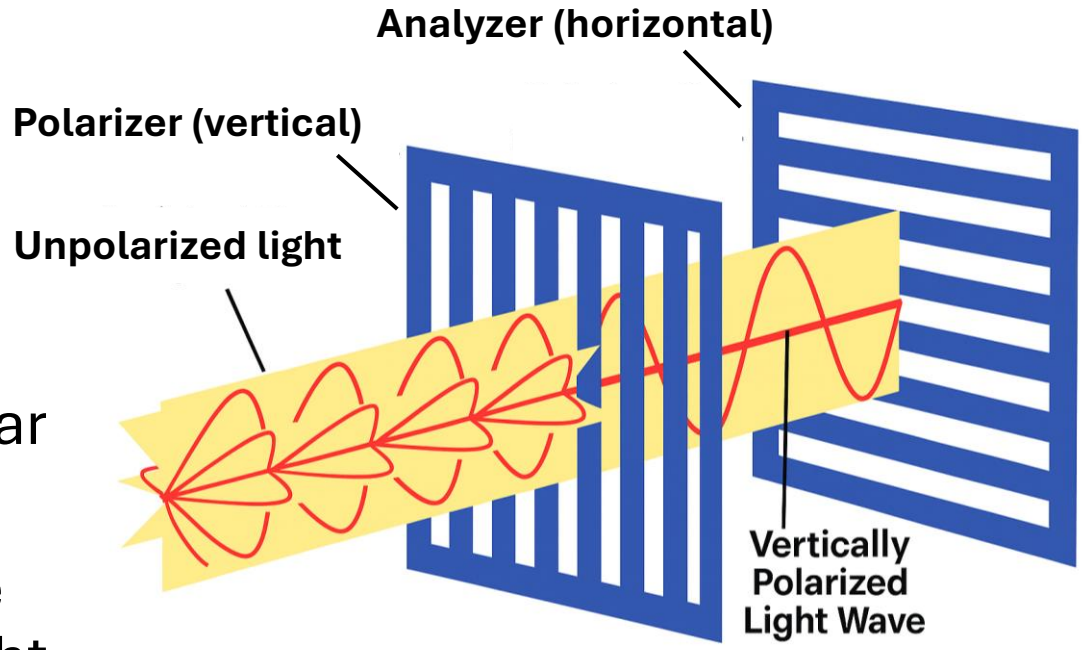
Color Filters

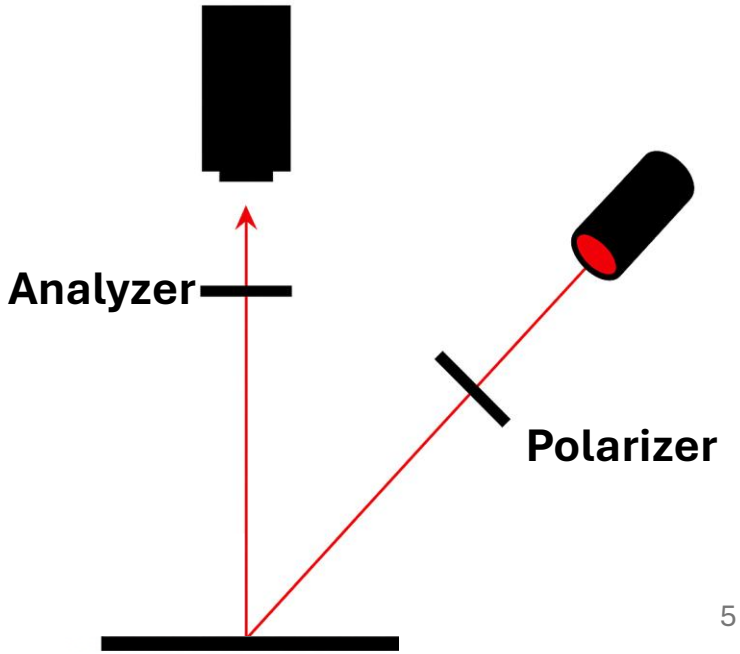
- A **color filter** is an optical tool or digital technique that allows specific wavelengths (colors) of light to pass through while blocking or suppressing others.
- It is commonly used to improve image contrast, reduce background interference, or highlight specific features, especially in **color-based object detection** and **visual inspection**.



Polarizing Filters

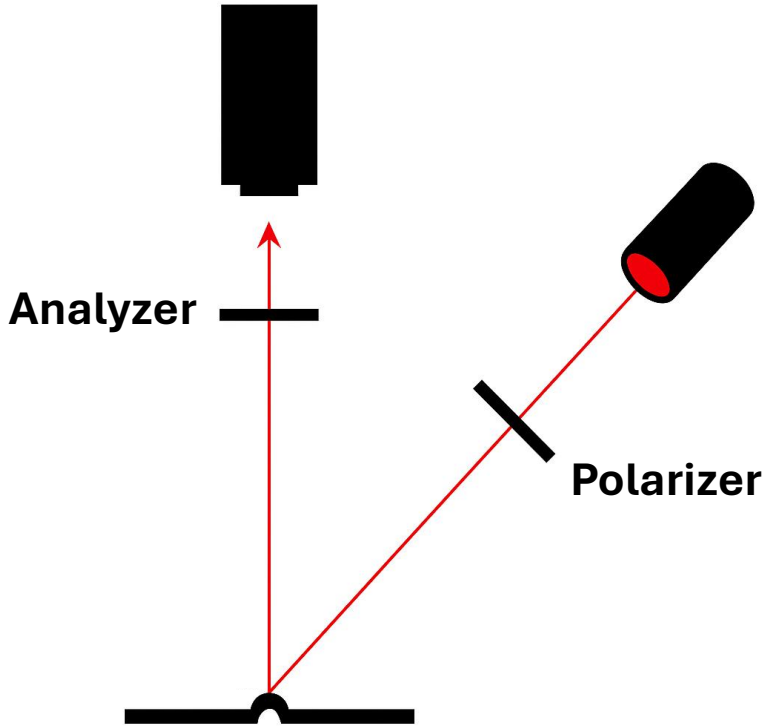
- A **polarizing filter** works like slatted blinds, allowing only light waves oscillating in a particular direction to pass through.
- Two filters are used together. The first, called the **polarizer**, polarizes the incoming light. When light reflects from a shiny, flat, non-metallic surface, the reflected light often becomes strongly polarized.
- The second filter, called the **analyzer**, is rotated so its transmission axis is perpendicular to the dominant polarization direction of the reflected light. **This blocks much of the reflected light**, reducing glare and making the underlying features easier to see.





Polarizing Filters

- If the part's surface is rough or uneven, the reflected light scatters in different directions. Some of this scattered light may pass through the analyzer and reach the camera.
- By blocking much of the strongly polarized reflection from shiny surfaces, **crossed polarizers reduce glare and hot spots, improve image contrast, and make surface features easier to see.**

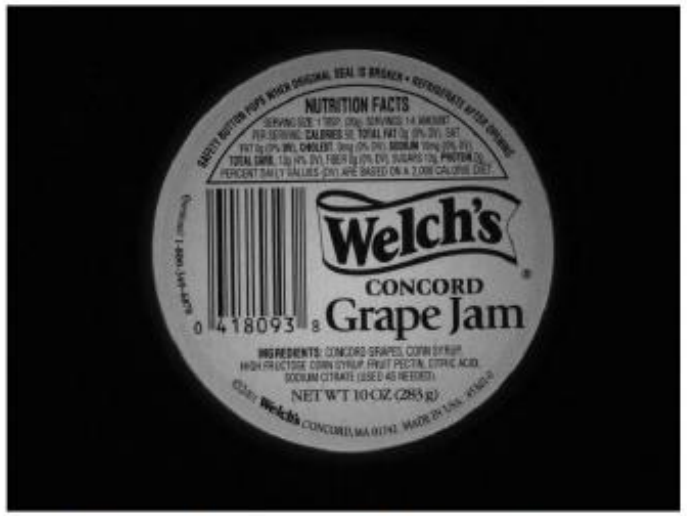


Polarized Light Application

Reading the Information on a Grape Jam Lid

- In the left image, direct lighting without a polarizing filter creates glare on the raised parts of the lid, making the information difficult to read. However, the information in the recessed areas remains visible.
- In the right image, adding a polarizing filter to the light source reduces the glare, **making the information across the entire lid clearly visible**.

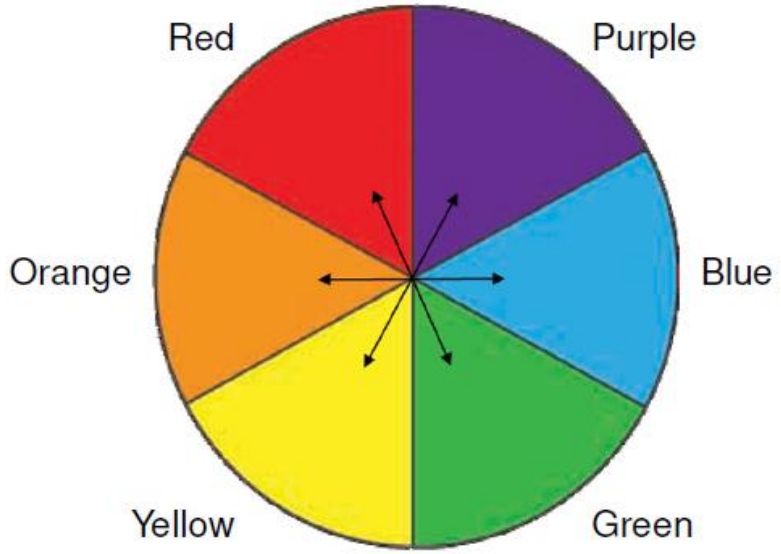




Colored Light

Using Colored Light with Grayscale Cameras

- Grayscale cameras capture brightness rather than color. By selecting the appropriate light color, we can control how different features appear in the image, making them stand out or fade away.
- **Like colors brighten features**: When the color of the light matches the color of the object on the color wheel, the object generally appears brighter. Example: Yellow light makes yellow objects appear brighter.
- **Opposite colors darken features**: When the light color is opposite to the object's color on the color wheel, the object generally appears darker. Example: Green light makes red objects appear darker.



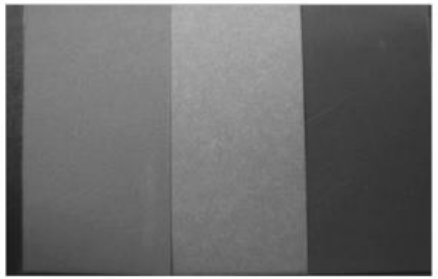
When illuminated by a light of the opposite color on the color wheel, an object reflects less light back to the camera, causing it to appear darker in the grayscale image

Colored Light Application

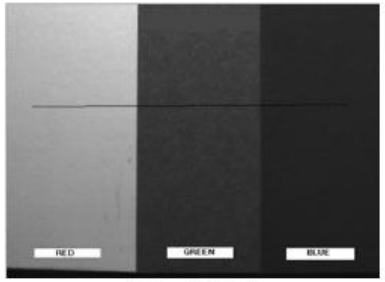
- In the **color image**, there are three strips: **red, green, and blue**. In the **grayscale image**, these colors appear as different shades of gray.
- If someone asks, “Which one is green?”, you can easily identify it as the **middle strip** by looking at the color image.
- But what if you only have the **grayscale image** and don't know the original order of the colors?
- We can use different **colored lights** to help identify them. Under **red light**, the red strip will appear **brighter**. Under **green light**, the green strip will appear brighter. Similarly, under **blue light**, the blue strip will appear brighter.



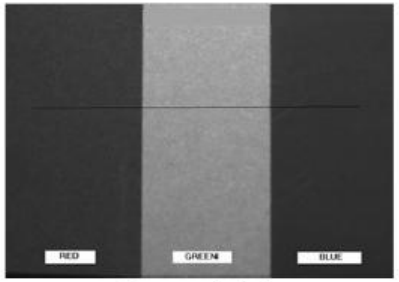
Actual Colors



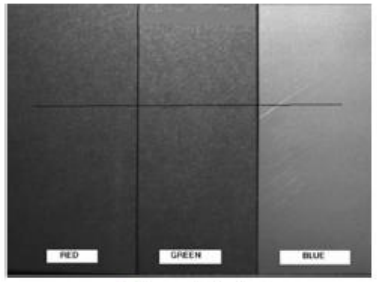
White Light



Red Light



Green Light



Blue Light

Colored Light Application

- In this application, we want to **enhance the date code** on a jar lid while **suppressing the other printed lettering**.
- Since we are interested in the **red print** and want to eliminate the **blue print**, we can use **blue light**.
- Notice how the blue print blends into the background, while the red date code stands out clearly.
- The **date information is accentuated** under blue light.
- The **blue print is suppressed**, while the **red print appears darker**, making the date code easier to detect.



Ambient Light



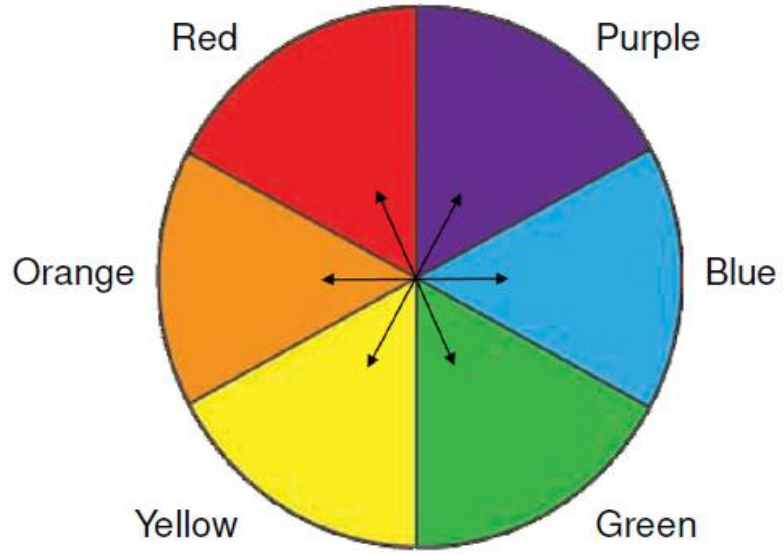
White Light



Blue Light

Colored Light

- What colored light would you use to transform the left image into the right image?

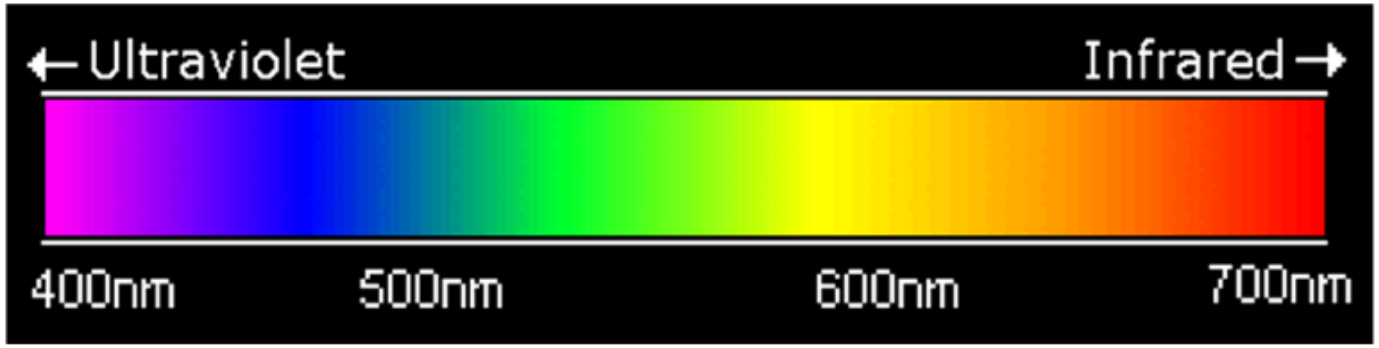






Infrared Light (700 nm – 1 mm)

- **Infrared (IR) light** is **invisible** to the human eye.
- Operators may not realize that IR light is being used unless they view the scene through an **IR-sensitive camera**.
- **IR imaging removes color information**, so the resulting image appears in shades of gray.
- IR light generally does not pass through solid materials, but it can penetrate **slightly below the surface of some soft or organic materials**, such as fruit skin, to reveal hidden features.



Infrared Light (700 nm – 1 mm)

Common uses:

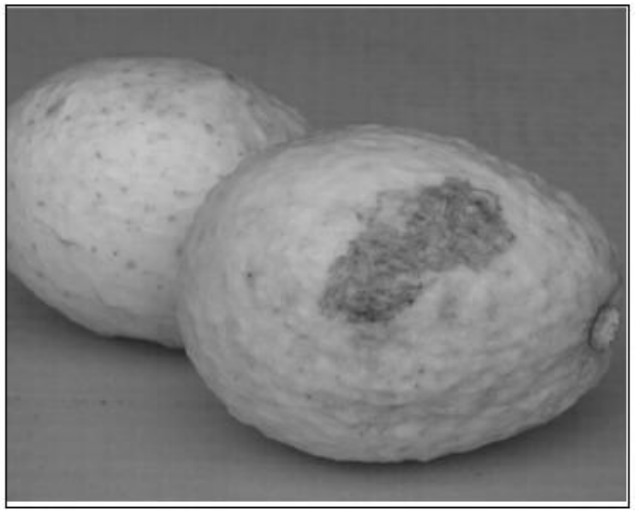
- **Barcodes and Shrink-Wrap:** IR light can penetrate thin plastic films like shrink-wrap, allowing barcode scanners to read labels without removing the packaging.
- **Livestock Monitoring:** IR cameras are used to monitor body temperature and detect health issues in animals, such as inflammation or infection, by spotting heat patterns.
- **Color Crayons:** Different pigments in crayons reflect IR light differently. This can be used to identify or sort colors even when they look similar in visible light.

Infrared Light Application

- In this application, an avocado may look perfectly good under **normal ambient light**, but when illuminated with **IR light**, a hidden bruise becomes visible.
- This is because IR light can penetrate slightly below the fruit's skin and may reflect differently in **damaged areas** compared with healthy areas, revealing hidden bruises.
- When using IR illumination, an appropriate **IR-pass filter** should be used on the camera to block visible light and ensure that the camera primarily captures **infrared light**.



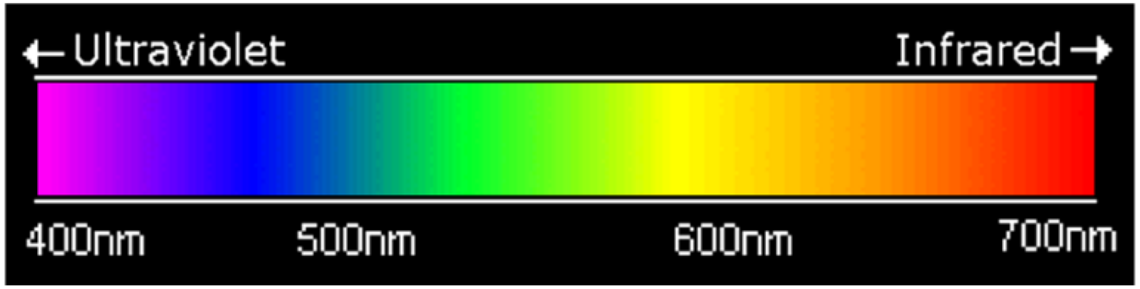
Ambient Light



IR Light and IR Filter

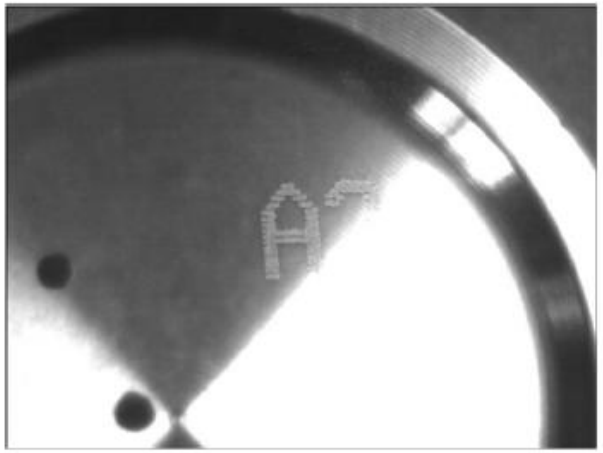
Ultraviolet Light (10 – 400 nm)

- **Ultraviolet (UV) light** can cause certain materials, such as ink, labels, or glue, to **fluoresce**. Fluorescence occurs when a material absorbs UV light and emits **visible light**. In applications, the visible light produced by fluorescence is what matters, not the UV light itself.
- When UV light shines directly on a part, the **fluorescent areas glow**, making them visible to the camera.
- A suitable **filter** is then used to block the UV light while allowing the **visible fluorescent light** to pass through to the camera.



Ultraviolet Light Application

- When we view the part under **UV light without a filter**, the ink used for the numbers fluoresces, making the text visible. However, the camera also captures **unwanted glare from the UV light source**.
- When we add a suitable **UV-blocking filter**, the UV light is blocked while the **visible light emitted by the fluorescent ink** passes through to the camera. This reduces unwanted glare and makes the numbers easier to see.

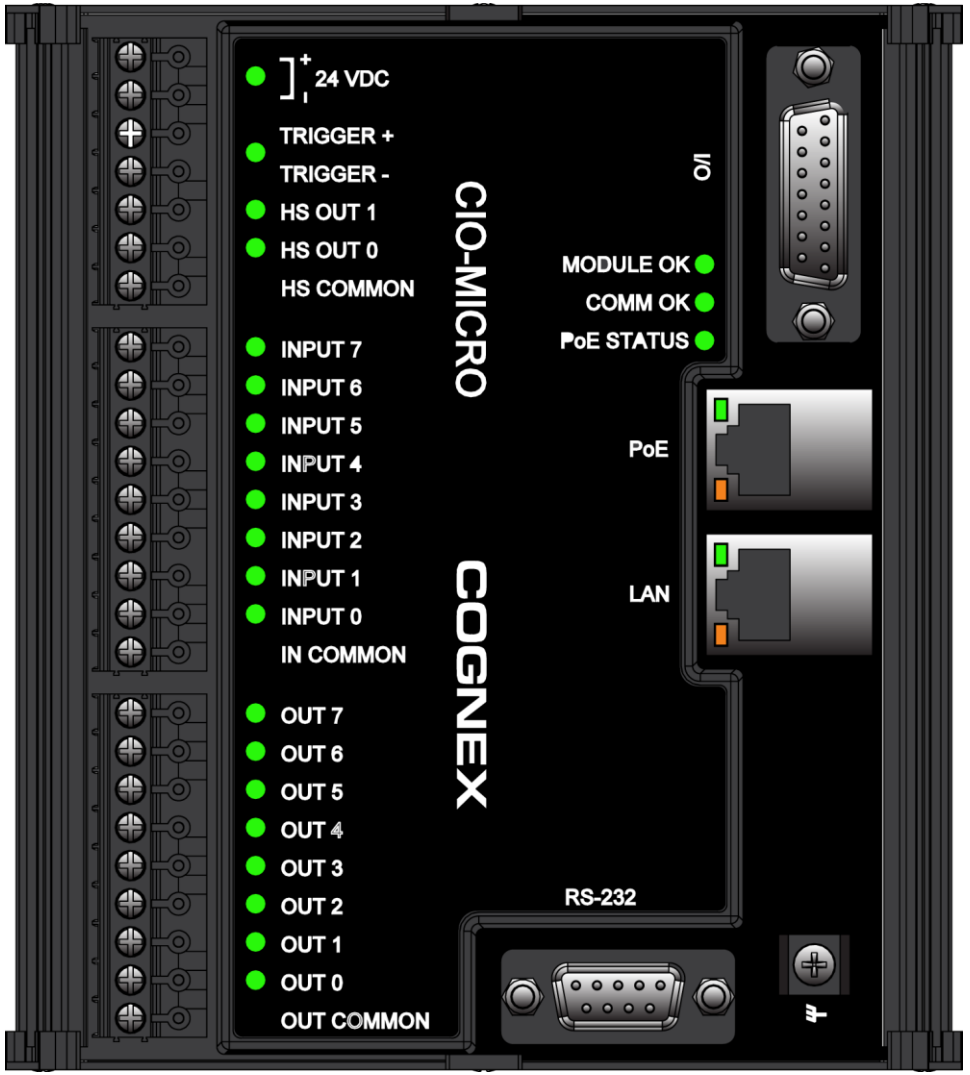


No Filter



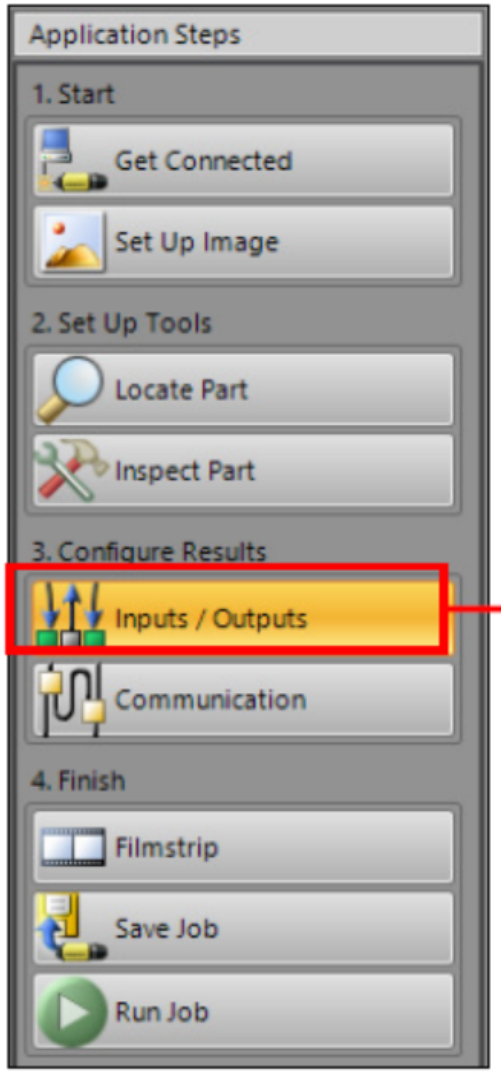
Longpass UV Filter

Inputs/Outputs

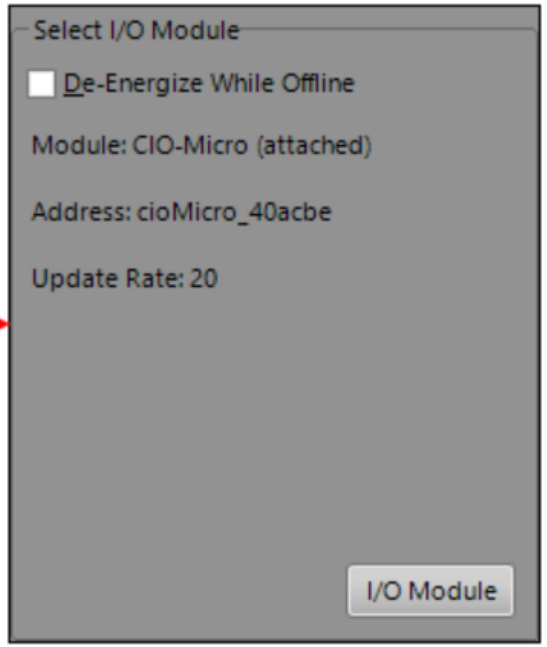


Inputs/Outputs

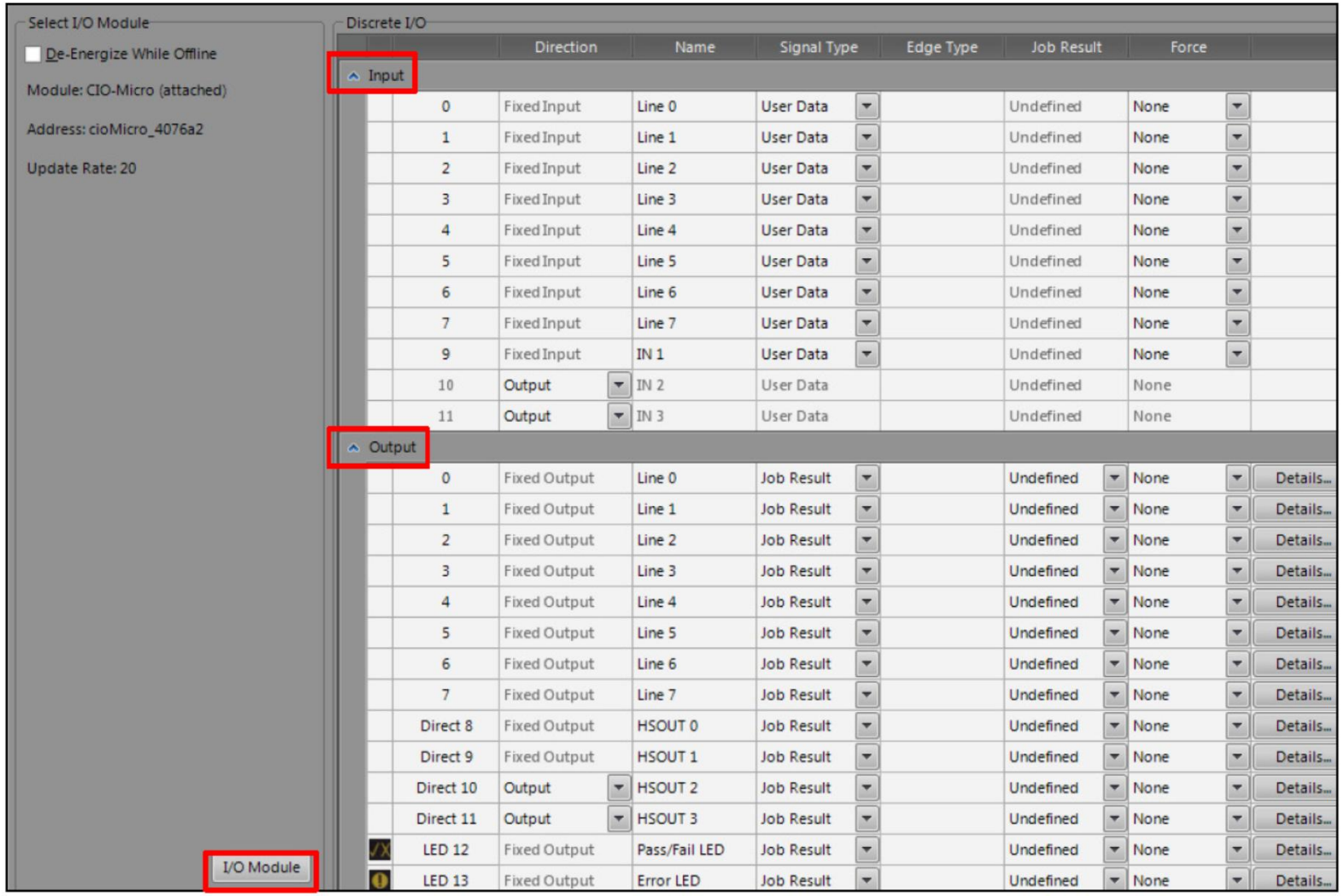
- The **Inputs/Outputs (I/O) step** is used to configure the **discrete input and output lines** of supported In-Sight vision systems and I/O modules.
- You can customize the **line name**, set the **signal type**, select the **edge transition** and **tool results** that trigger a signal, and **force inputs or outputs** to test your application.



NOTE: *This application step is disabled when using an emulator.*

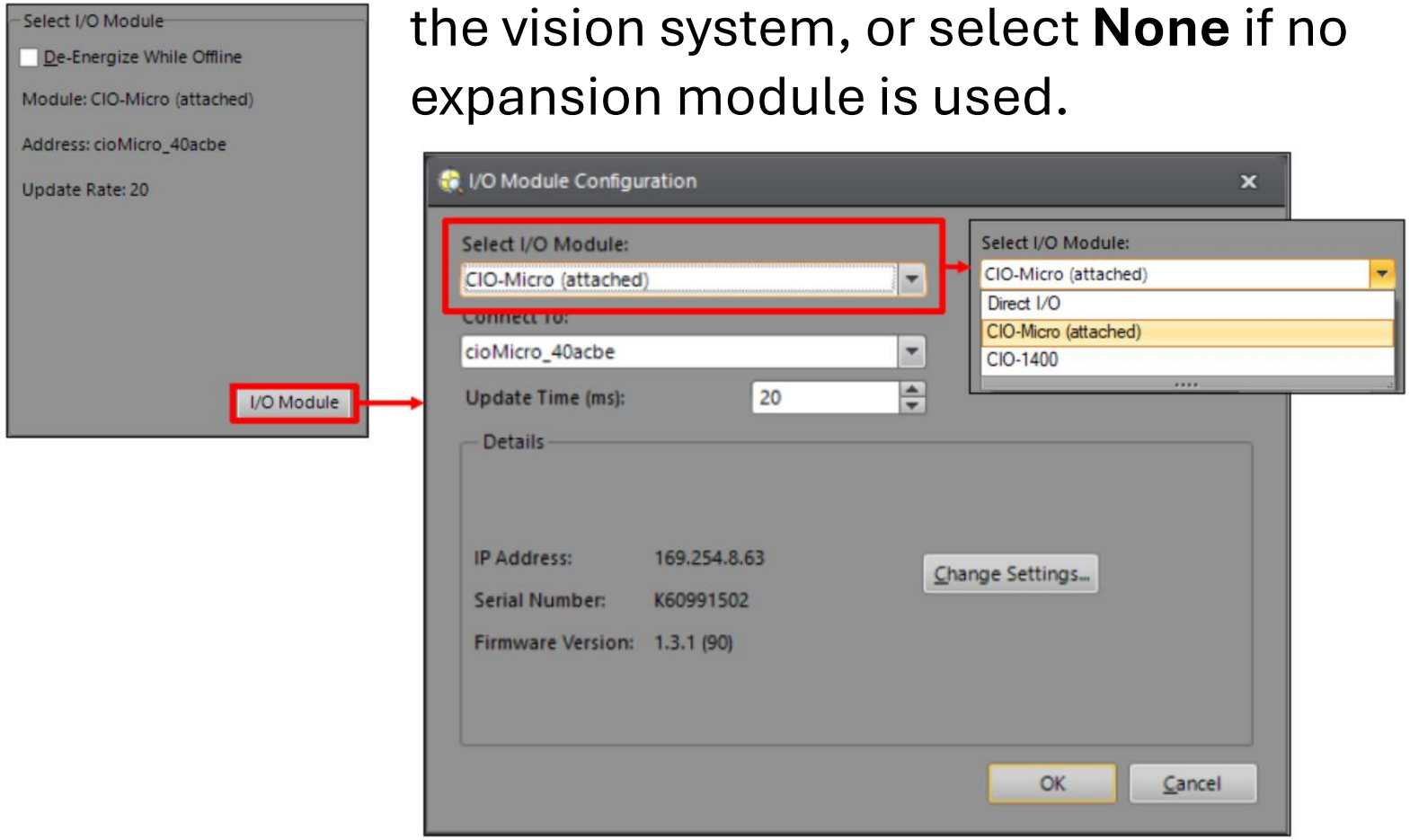


Select I/O Module



Select I/O Module

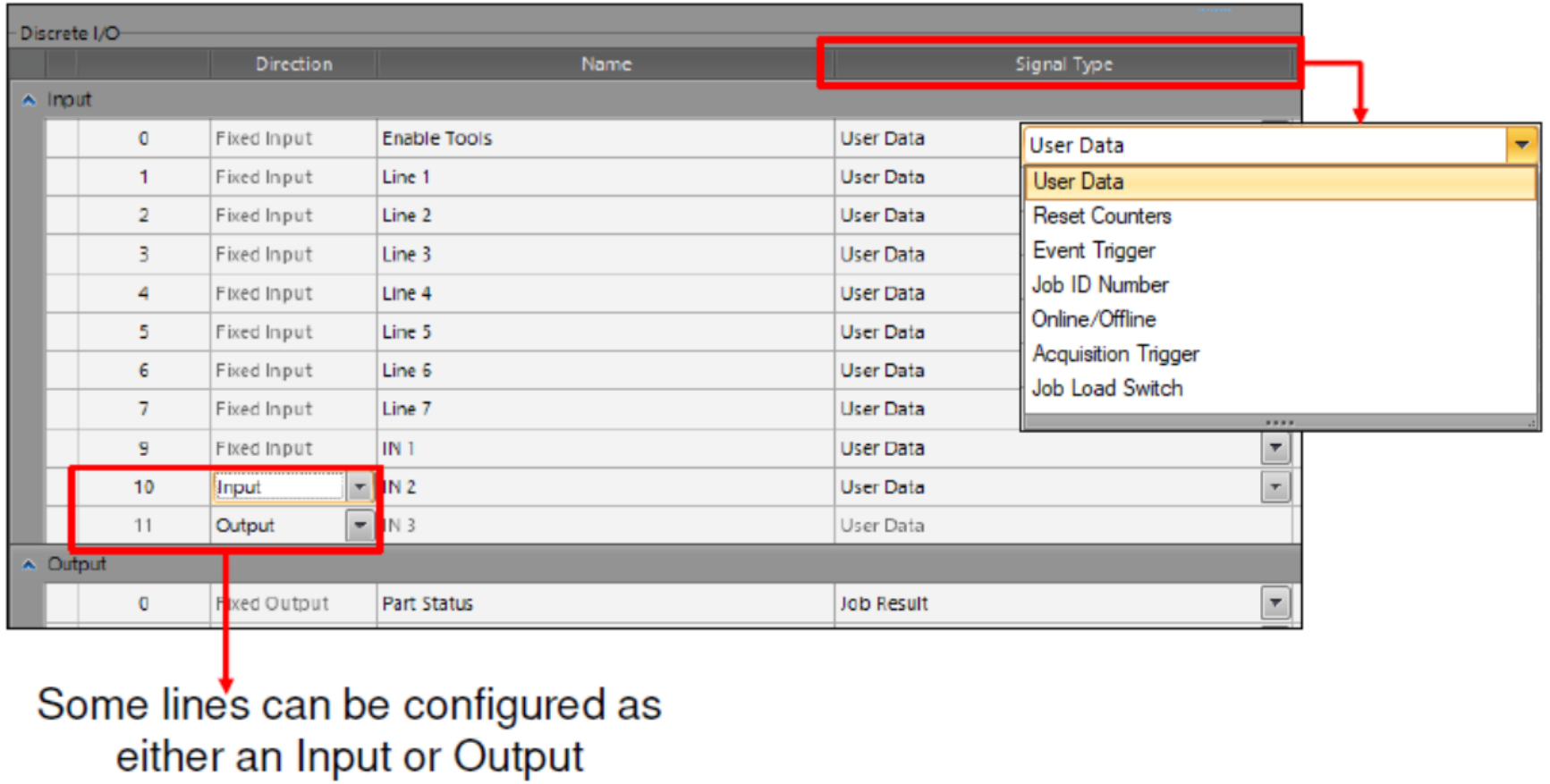
- The **I/O Module** button allows you to select the **I/O Expansion Module** being used with the vision system, or select **None** if no expansion module is used.



- **Direct** means that no **I/O Expansion Module** is linked to the camera.
- Only **one camera at a time** can be linked to a given **I/O Expansion Module**.
- To link a camera to a module that is already linked to another camera, the existing camera must first release the connection by setting its **module type to Direct**.

Select I/O Module - Inputs

- The **Discrete Inputs** settings dialog is used to configure the **discrete input lines** on the active In-Sight vision system.
- Each input line can be configured for a specific function, such as:
- **User Data** – A general-purpose input used to turn **Location and Inspection tools On or Off**.

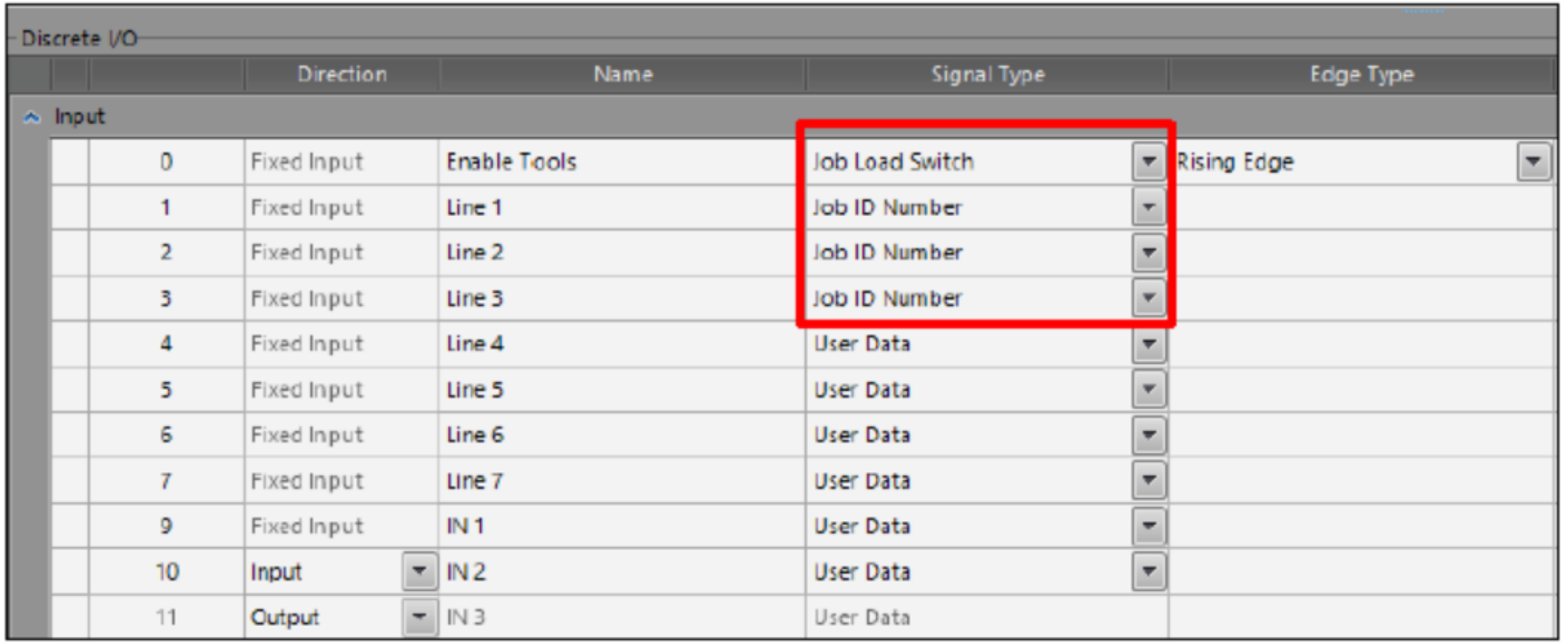


Select I/O Module - Inputs

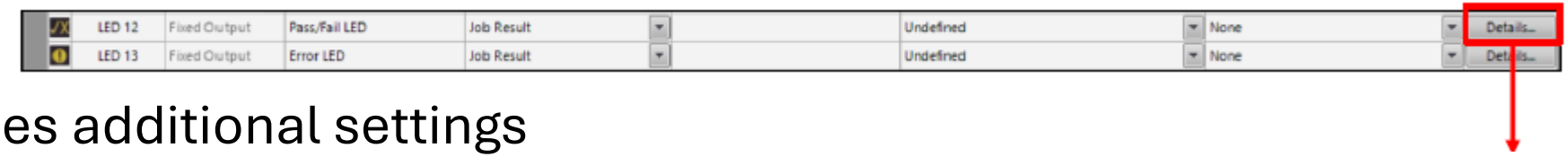
- **Reset Counters** – Resets the EasyBuilder counters (Job.Fail_Count, Job.Inspection_Count, Job.Pass_Count, <Tool>.Error_Count, <Tool>.Fail_Count, and <Tool>.Pass_Count) to **0**.
- **Event Trigger** – Triggers an event through logic created in Spreadsheet View. For example, an operator can press a button to save the current image.
- **Job ID Number** – Provides one bit of the **Job ID number**, which is used to select a job to load when a separate input line configured as a Job Load Switch is ON.
- **Online/Offline** – Forces the vision system to go **Online or Offline**. **LOW (0) = Offline**, and **HIGH (1) = Online**.
- **Acquisition Trigger** – Triggers the vision system to **acquire an image**.
- **Job Load Switch** – When ON, reads all the **Job ID Number** input lines and loads the specified job.

Select I/O Module - Inputs

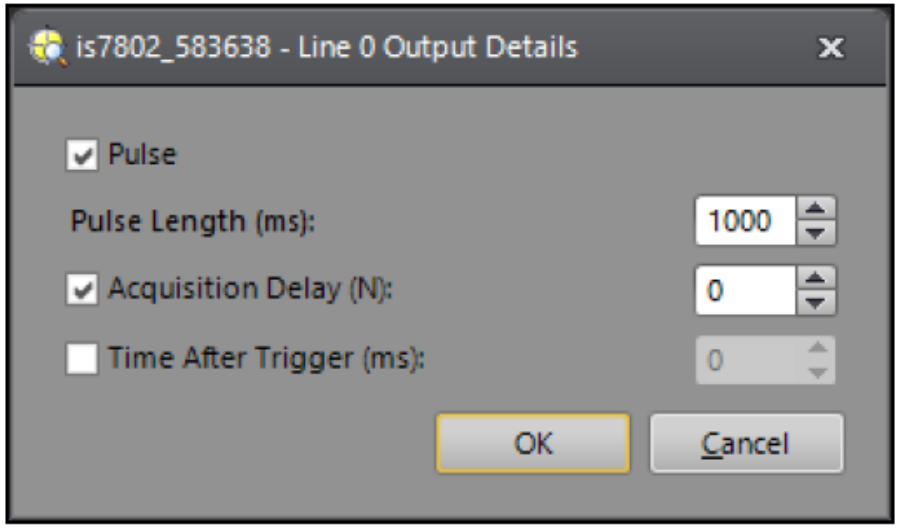
- When the **Job Load Switch** is activated, the vision system reads the **Job ID Number** input lines and loads the job corresponding to the specified binary number.
- **Examples:**
  000 – MyJob
  011 – NortheastLine
  101 – GearInspect



Select I/O Module - Outputs

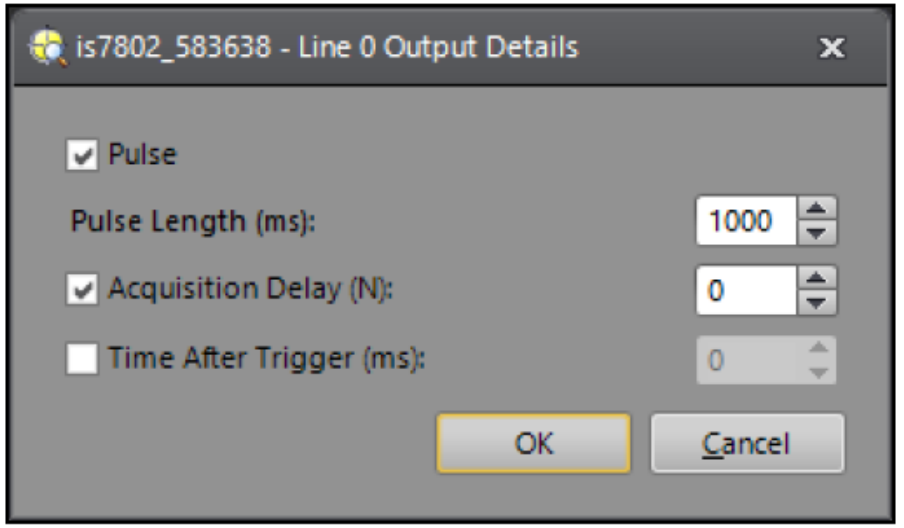


- The **Details** dialog provides additional settings depending on the selected output **Type**.
- **Pulse** – When selected, the output is activated for the specified **Pulse Length** and then deactivated. When cleared, the output remains in its current state until it is changed. The output must be pulsed when the **Acquisition Delay** is greater than 0.
- **Pulse Length (ms)** – Specifies the duration of the output pulse in milliseconds.
  In-Sight Micro 1000 series, In-Sight 5000, and In-Sight 8405: 10–1000 ms (default = 10 ms).
  In-Sight 7000 series and In-Sight Micro 1402, 1412, and 1500: 1–1000 ms (default = 10 ms).



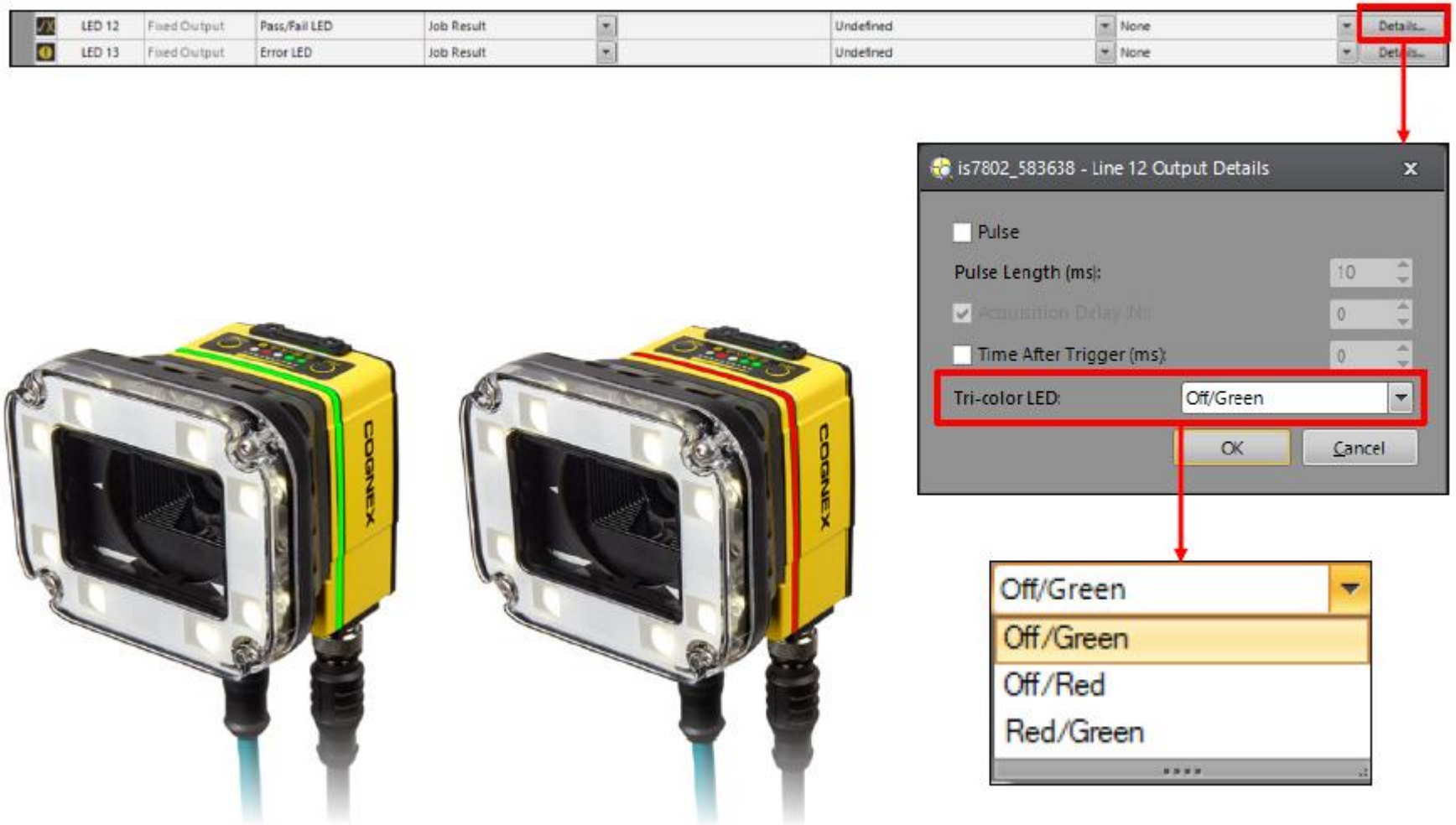
Select I/O Module - Outputs

- **Acquisition Delay (N)** – Specifies the number of acquisition or tracking pulses (0–1000) to delay the output after a signal pulse is received by an output line. If **Acquisition Delay = 0**, the output line is updated immediately when the WriteDiscrete function is evaluated. If **Acquisition Delay > 0**, the output line must be pulsed.
- **Time After Trigger (ms)** – When selected, the output is activated after the specified time delay (0–10,000 ms) following the trigger.



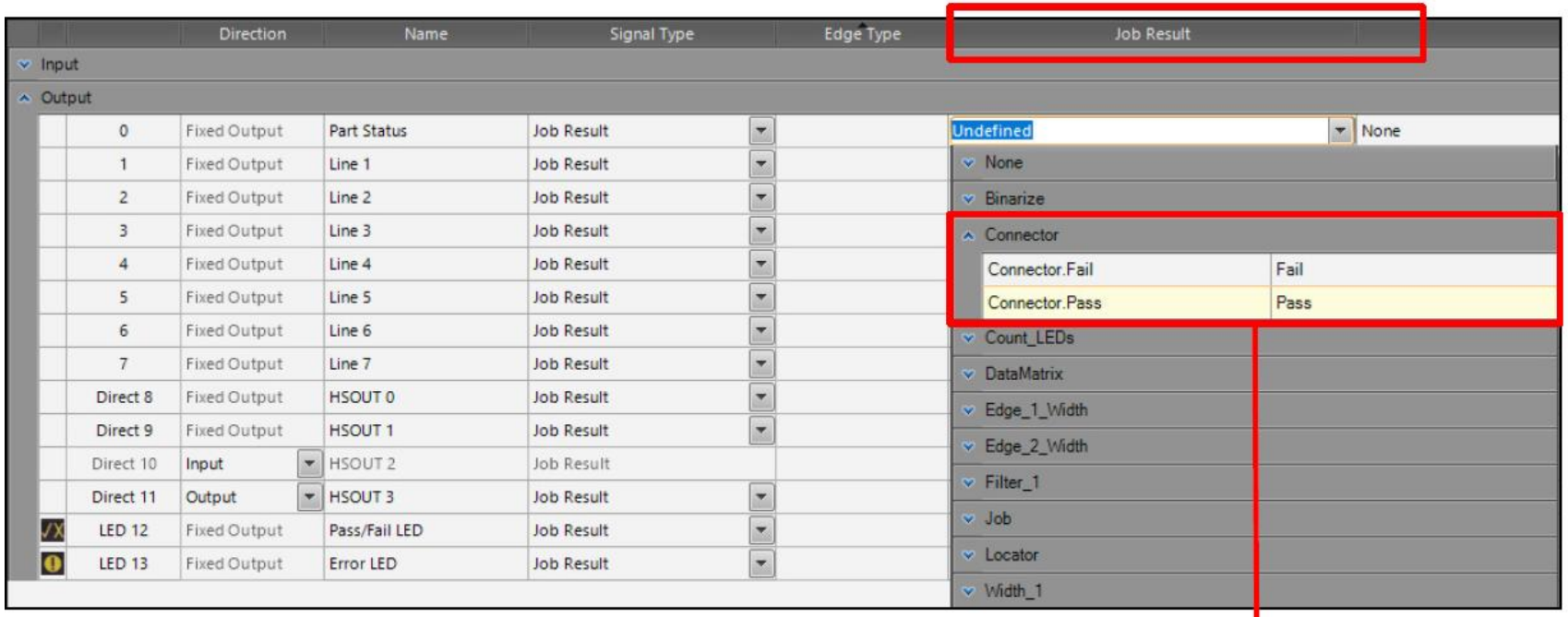
Select I/O Module - Outputs

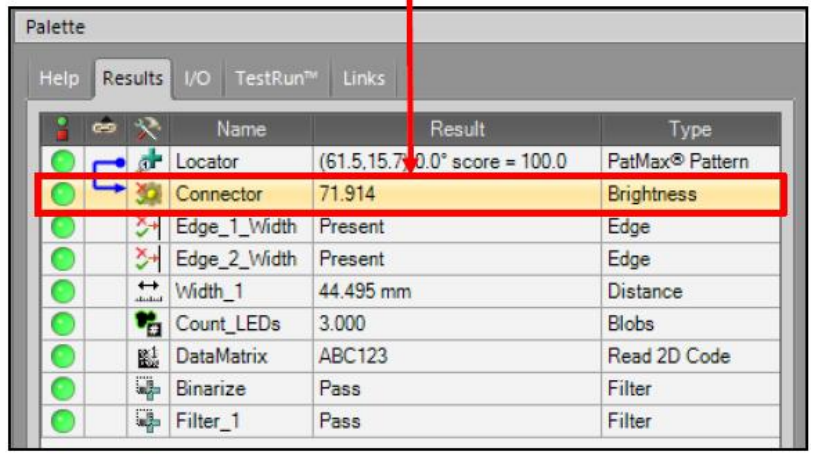
- **Tri-color LED**: specifies pass/fail colors on cameras with Tri-color LED, e.g., In-Sight 7802.
- **Off/Green**
  - Output OFF → LED OFF
  - Output ON → LED GREEN
- **Off/Red**
  - Output OFF → LED OFF
  - Output ON → LED RED
- **Red/Green**
  - Output OFF → LED RED
  - Output ON → LED GREEN



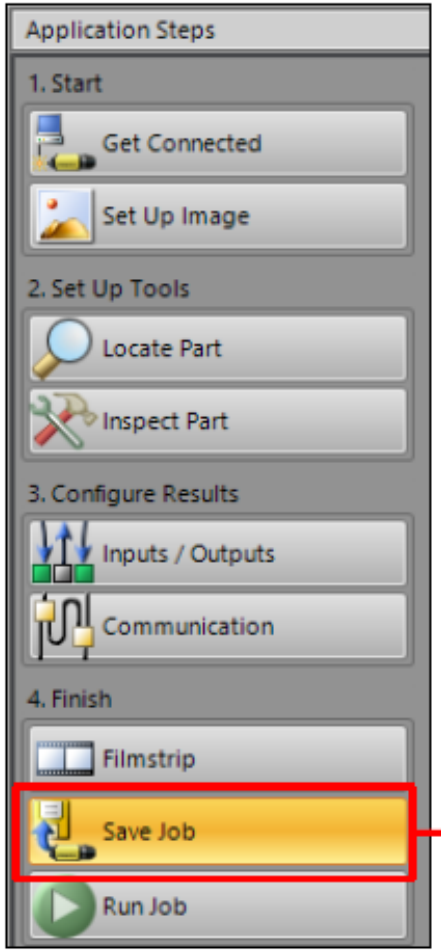
Select I/O Module - Outputs

- When an output is configured to report the pass/fail status of a tool, the output signal is **High (1)** when the tool result is **Pass** and **Low (0)** when the tool result is **Fail**.

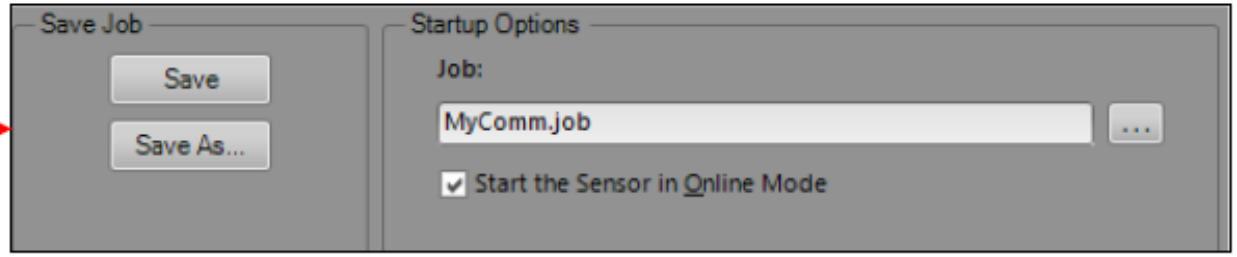




Save Job

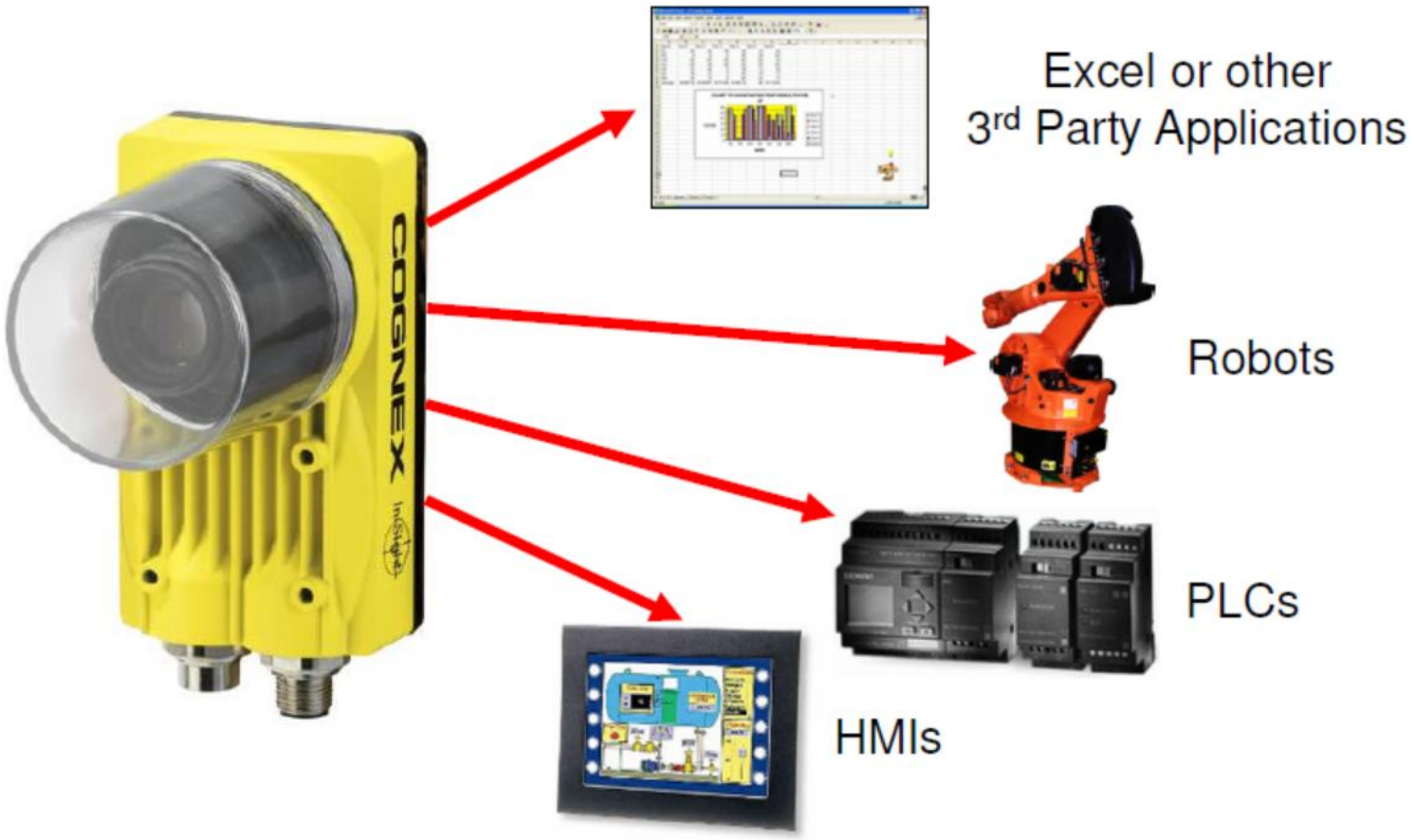


- The **Save Job** step allows you to perform various **file-management and sensor-maintenance tasks** while the camera is Offline.
- The **Startup Options** group box allows you to specify tasks that are **automatically performed when the In-Sight camera is powered on**, such as **loading a default job** and placing the camera *Online*.



Network Communications

Network Communications

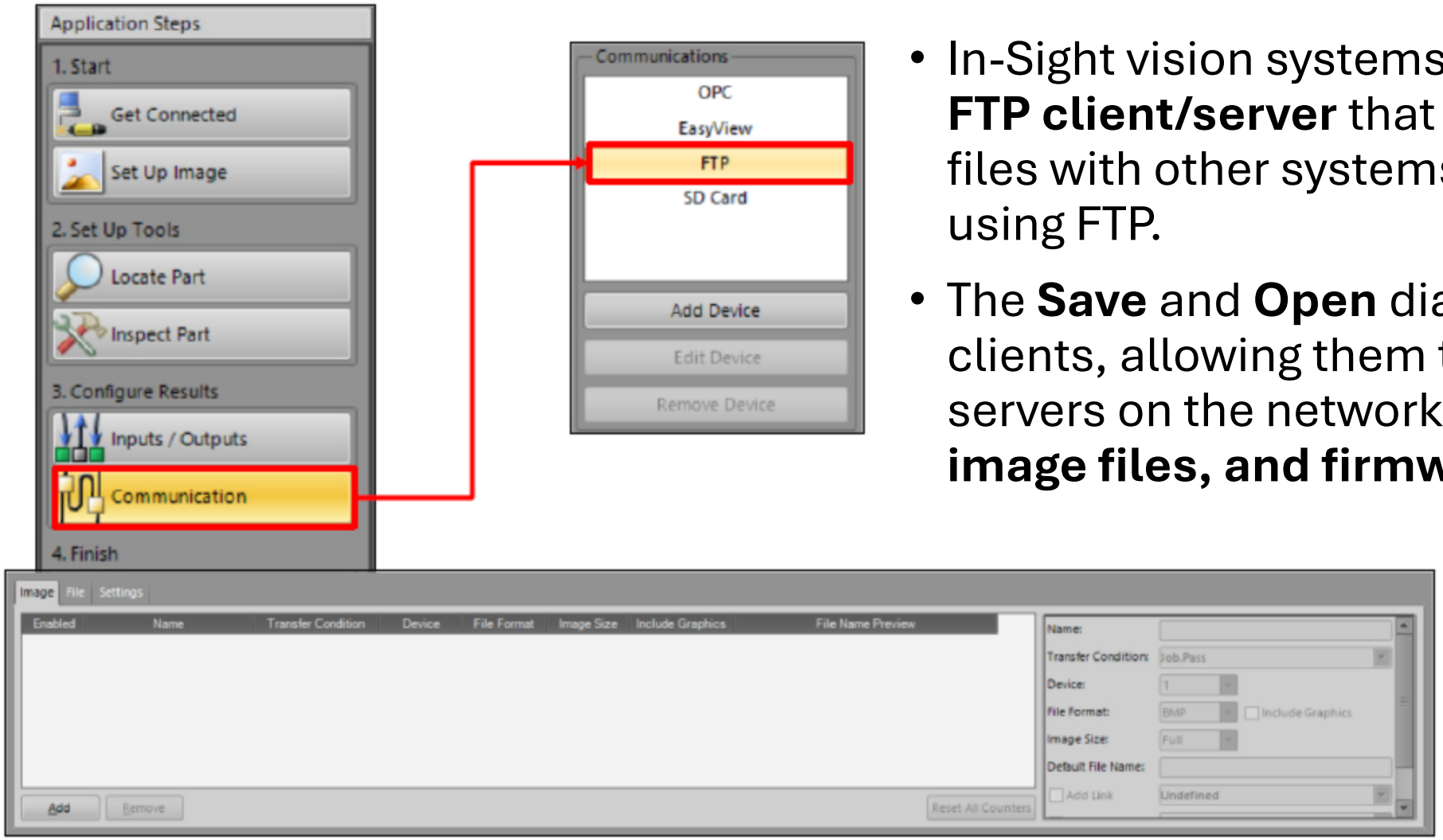


Network Communications

The **Communication** step is used to:

- Define inspection results for **Open Data Access and Exchange (OPC)**.
- Configure **EasyView** for the VisionView™ Operator Interface Panel and export images via **File Transfer Protocol (FTP) or to the SD card**.
- Configure **network and serial communication settings** so In-Sight vision systems can exchange data with other devices, such as a **robot, Programmable Logic Controller (PLC), or Motion Controller (MC)**.
- **NOTE:** The In-Sight camera must be **Offline** to enable the Communication step.

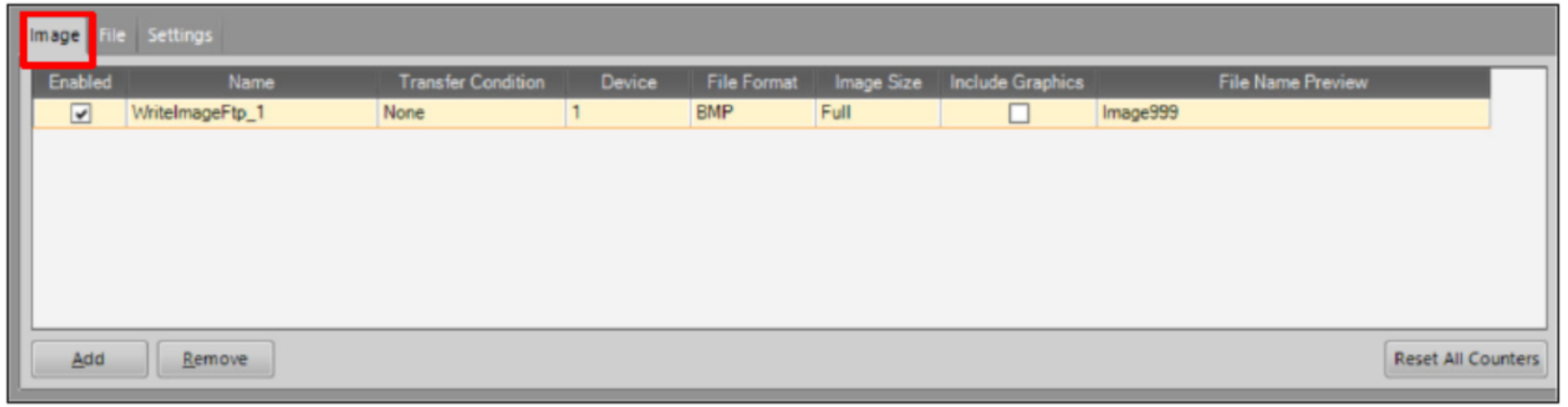
Network Communications

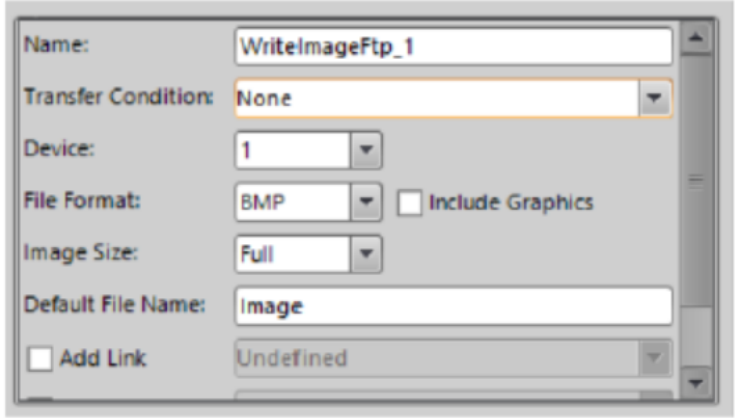


- In-Sight vision systems have an integrated **FTP client/server** that allows them to share files with other systems on the network using FTP.
- The **Save** and **Open** dialogs act as FTP clients, allowing them to connect to FTP servers on the network to transfer **job files, image files, and firmware update files**.

Network Communications

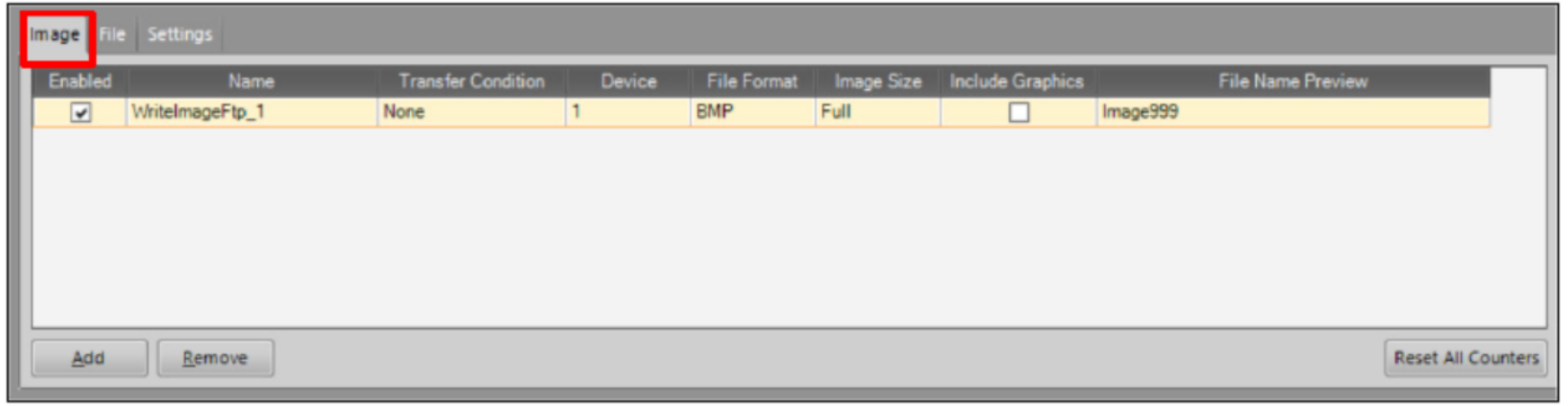
- On the **Image** tab, click the **Add** button. The **WriteImageFTP** tool appears.
- **Transfer Condition** defines **when image data should be sent to the FTP server**. For example, the transfer can occur when a particular tool passes or fails, or when an input is received.
- By default, no transfer condition is selected. When the **Enabled** checkbox is checked, the image is exported **every time an image is acquired**.
- If the **Enabled** checkbox is unchecked, or the tool referenced by the transfer condition is disabled, the image export operation will not execute.

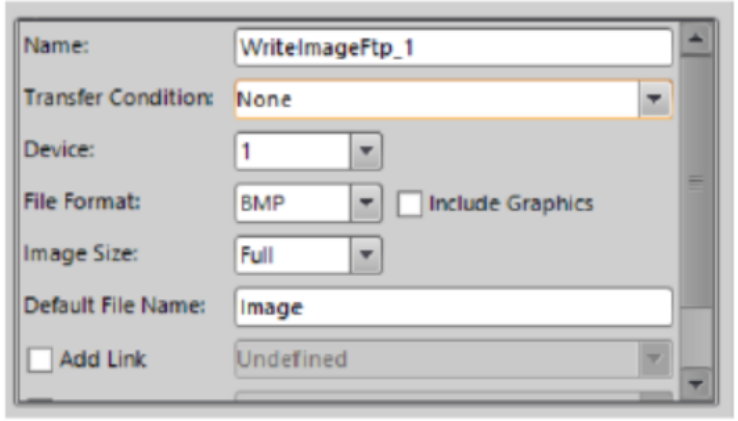




Network Communications

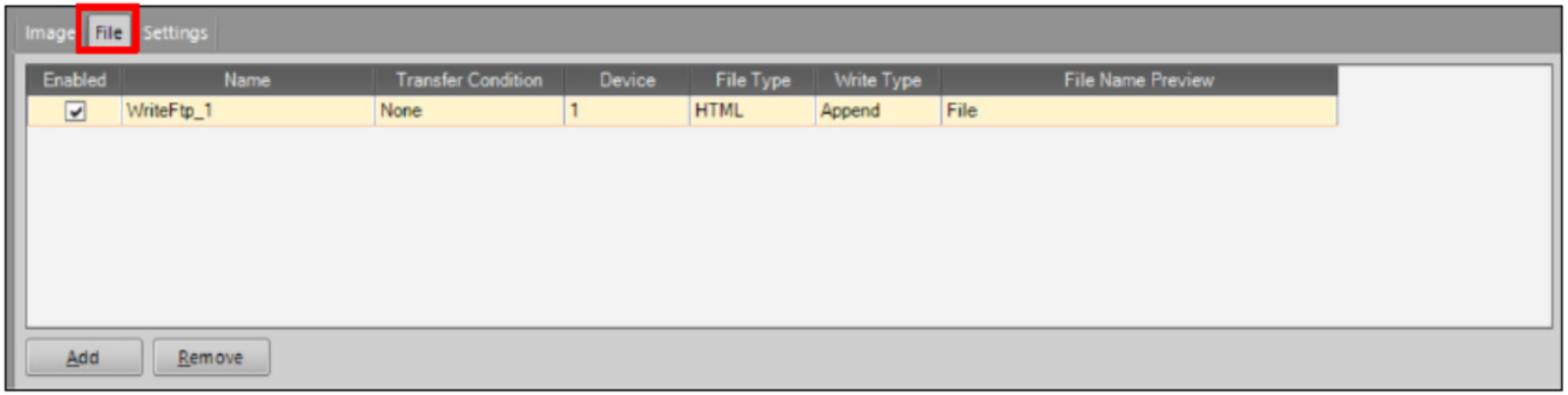
- **File Format** defines the file format of the exported image: **BMP** (Windows bitmap format) or **JPG** (JPEG format). By default, **BMP** is selected.
- **Default File Name** defines the name used for the exported image file. The default name is **Image**. If using In-Sight Explorer as the FTP directory, enter the authorized FTP directory followed by a backslash (\) and then the file name **Image**.

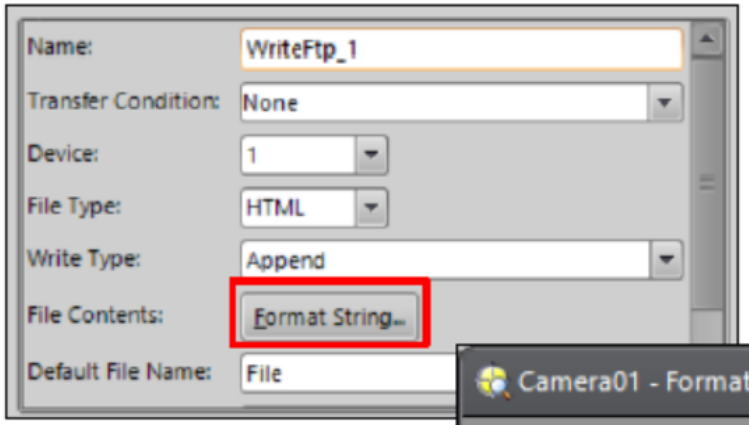


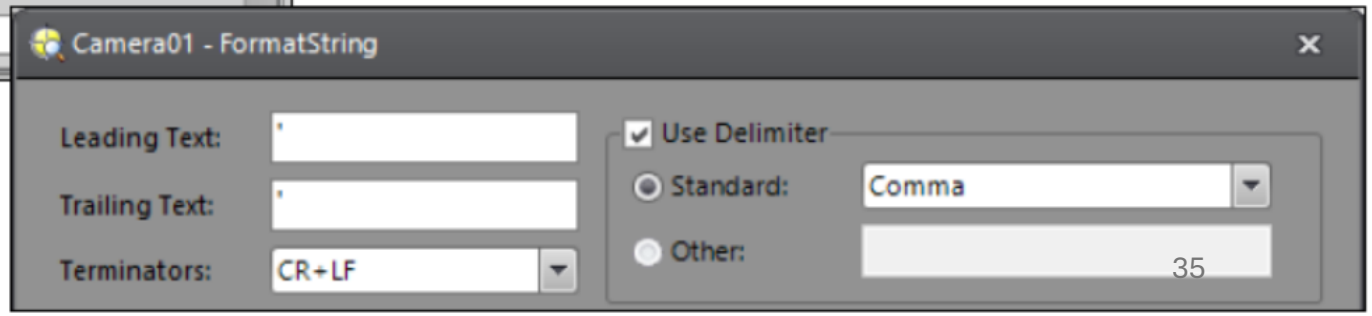


Network Communications

- On the **File** tab, click the **Add** button. The **WriteFTP** tool appears.
- **File Contents: Format String** defines the **data to be written to the file** using the FormatString dialog.

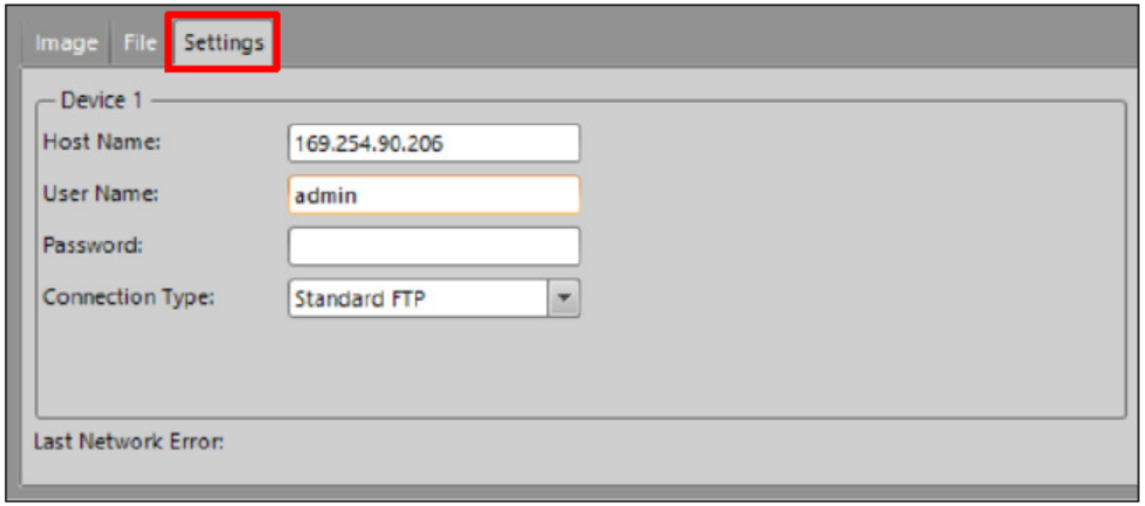






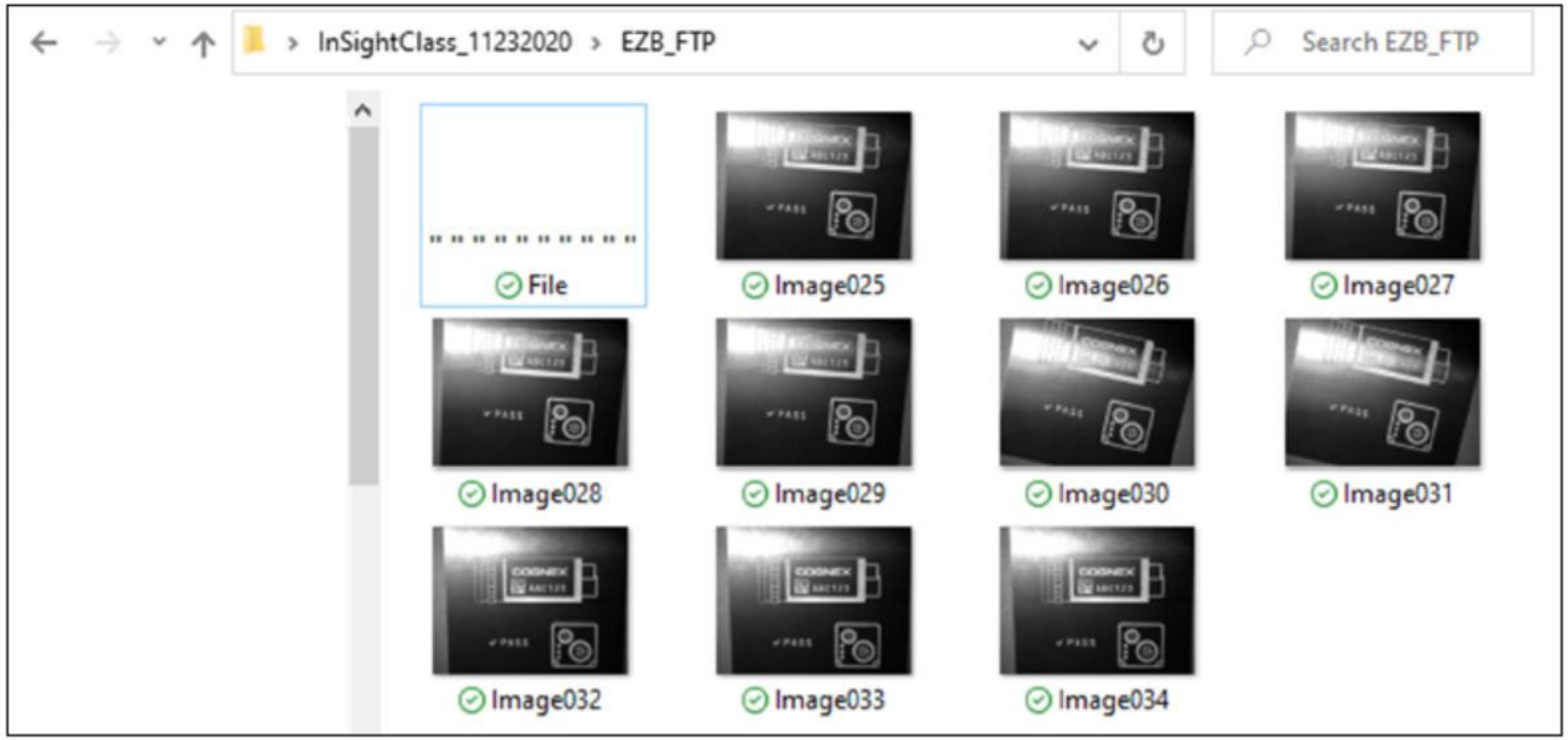
Network Communications

- The **Settings** tab is where you configure the In-Sight Emulator as the target device.
- **Host Name** defines the host name or IP address of the target device's FTP server.
- **User Name** defines the valid username for the target device's FTP server. This username does not have to exist on the In-Sight vision system that is exporting the image file or data.
- **Password** defines the valid password for the target device's FTP server. The password is **case-sensitive** and cannot exceed **20 characters**.
- Since we are using the In-Sight Emulator as the target device, the user name is **admin**, and the password is left blank.



Network Communications

- Place the camera **Online** and trigger it several times to capture images.
- Place the camera **Offline** and check the FTP folder. **A BMP file should be created for each captured image.**



Troubleshooting

Troubleshooting

The first question to ask is….

What has Changed since the last time the system worked?

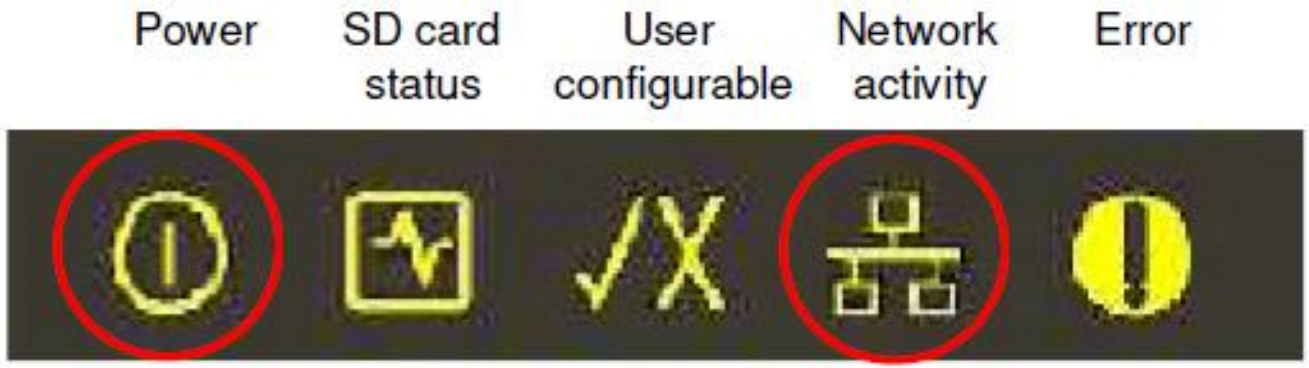
Troubleshooting

Use the following checklist as a guide to begin troubleshooting.

- Start with the **Hardware**, then move on to the **Software**.
- **Hardware:** Has anyone made changes to the hardware?
  - Cables rewired or connections changed.
  - Lighting changed.
  - Camera mounting adjusted.
- **Software:** Has anyone made changes to the software?
  - Parameters modified or fine-tuned.
  - A new program or job loaded.
- Is it possible to restore the system to its **original, known-good condition**?

Troubleshooting

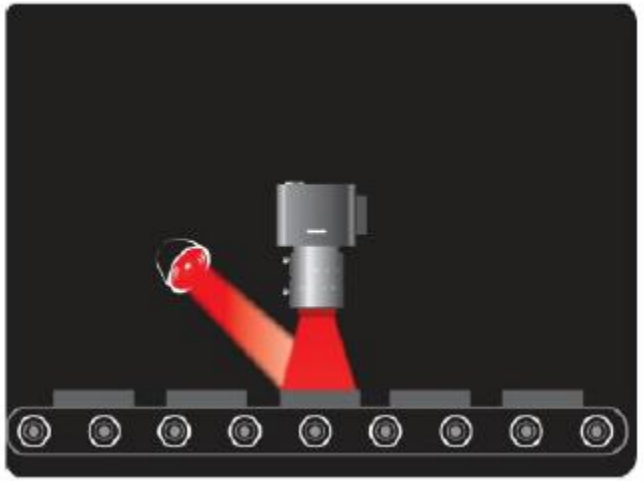
- Check that **power is supplied to the system** and the **network connection is established**.
- **2000 Series and 7000 Gen II Series:** The **Power LED** should be green, and the **Network LED** should be yellow.

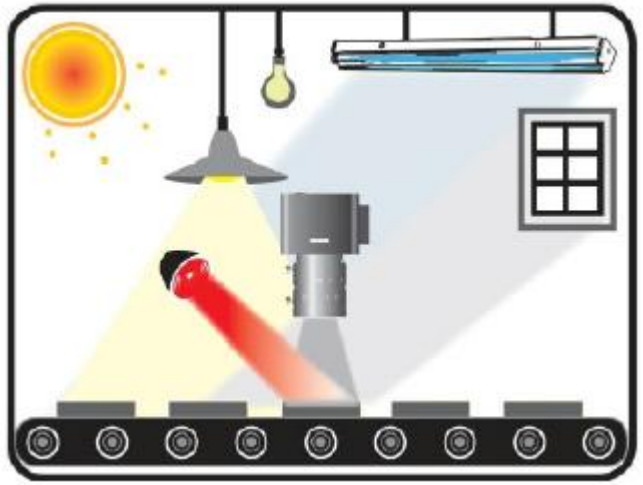


In-Sight 2000 Series and 7000 Gen II Series

Troubleshooting

- **Lighting and Optics** can cause a vision system to fail.
- **Lighting:** Is the lighting setup correct?
  - **Location and orientation** of the light source.
  - **Intensity** of the light.
  - **Color** of the light.
  - **Ambient light** affecting the image.
- **Lens:** Is the lens properly mounted and configured?
  - **Focus:** The image is out of focus.
  - **Aperture:** The aperture setting is incorrect.
  - **Lens cleanliness:** Debris or dirt on the lens.



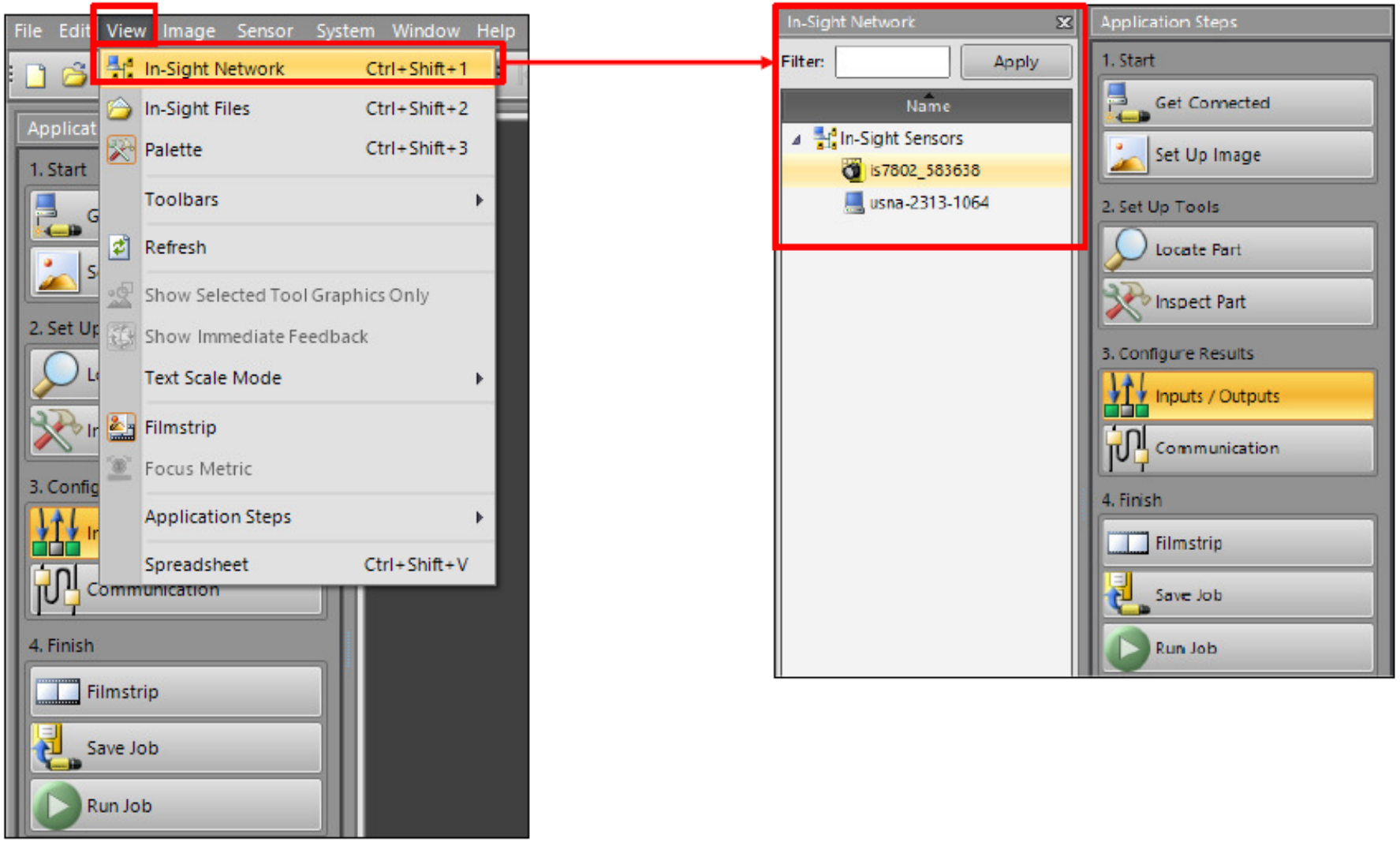


Troubleshooting

- A vision system may fail if it is **dirty or requires maintenance**.
- **Cleaning the Sensor Housing:** Remove dust and dirt from the exterior of the sensor housing.
- **Cleaning the CCD Window:** Carefully clean the CCD window to remove dust, dirt, and smudges that could affect image quality.

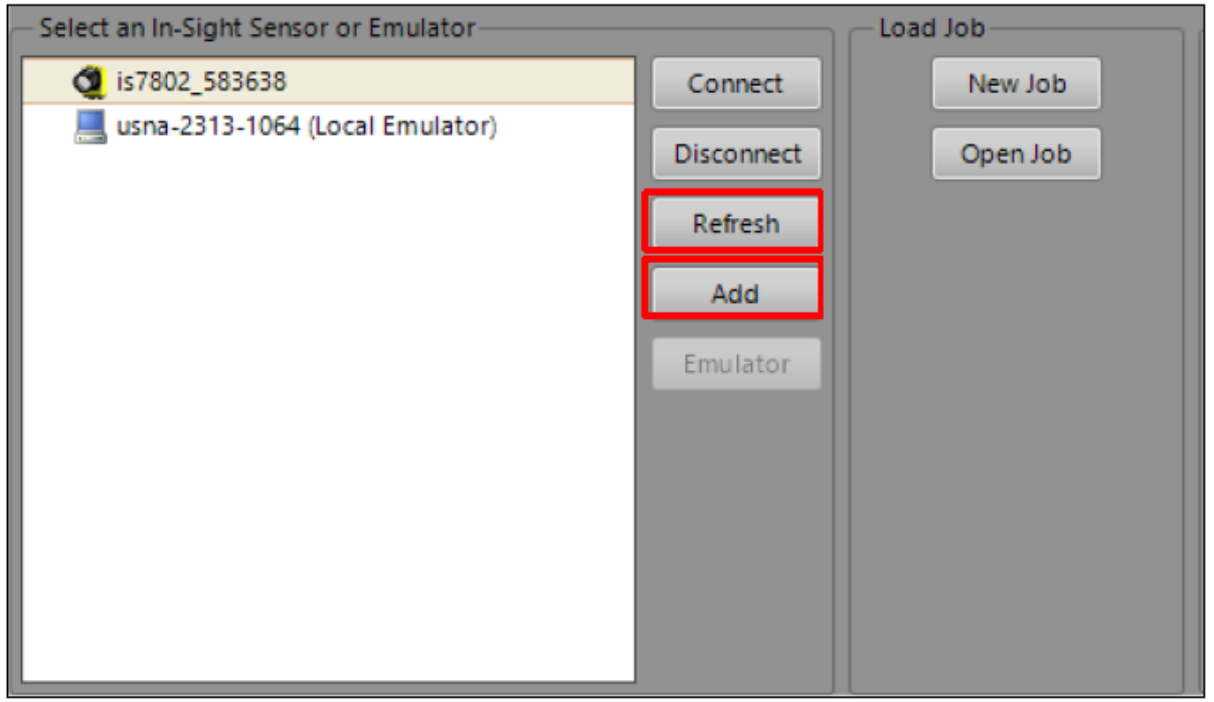


Network Troubleshooting



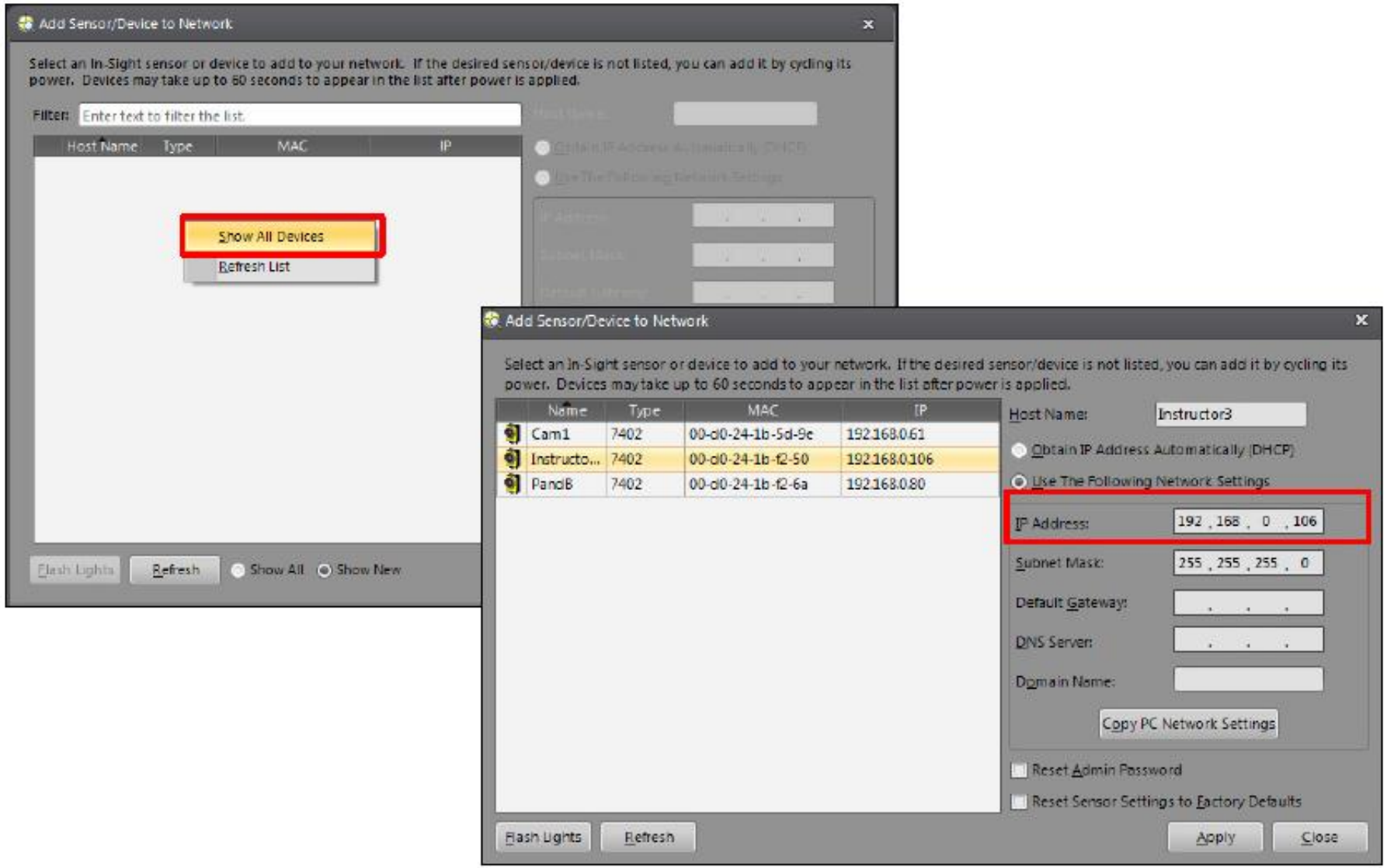
Network Troubleshooting

- Click the **Refresh** button to ensure that all In-Sight units on the network are detected.
- If the In-Sight sensor does not appear in the list, click the **Add** button to add it manually.



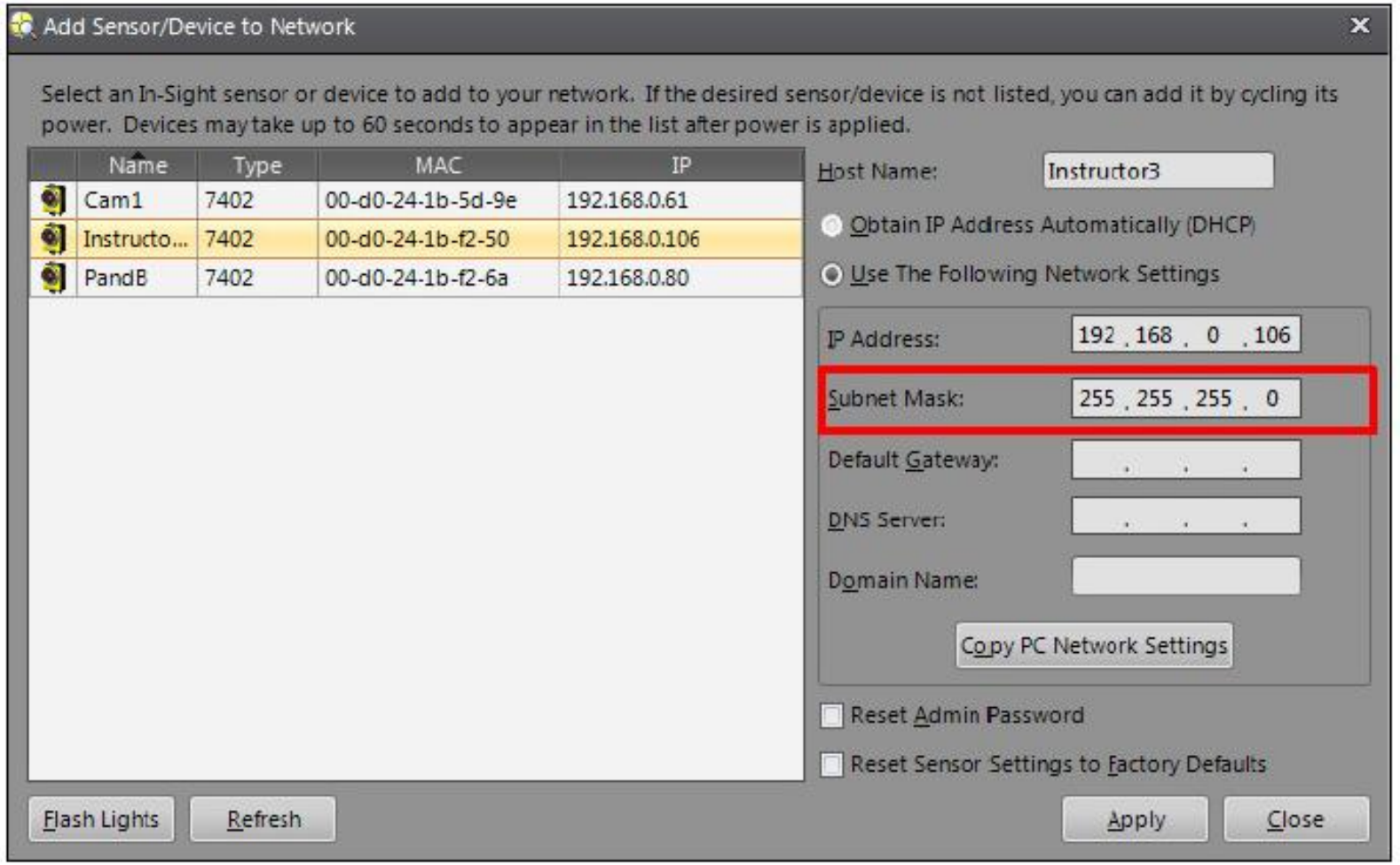
Network Troubleshooting

- Ensure that the sensor's **IP address** is configured correctly.



Network Troubleshooting

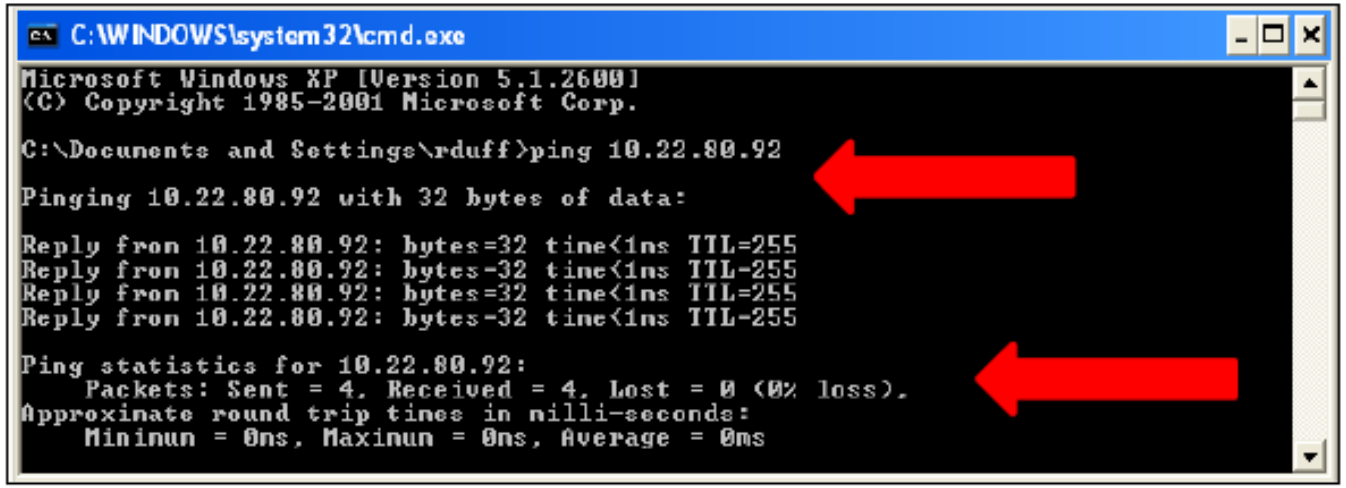
- Ensure that the sensor's **Subnet Mask** is configured correctly.



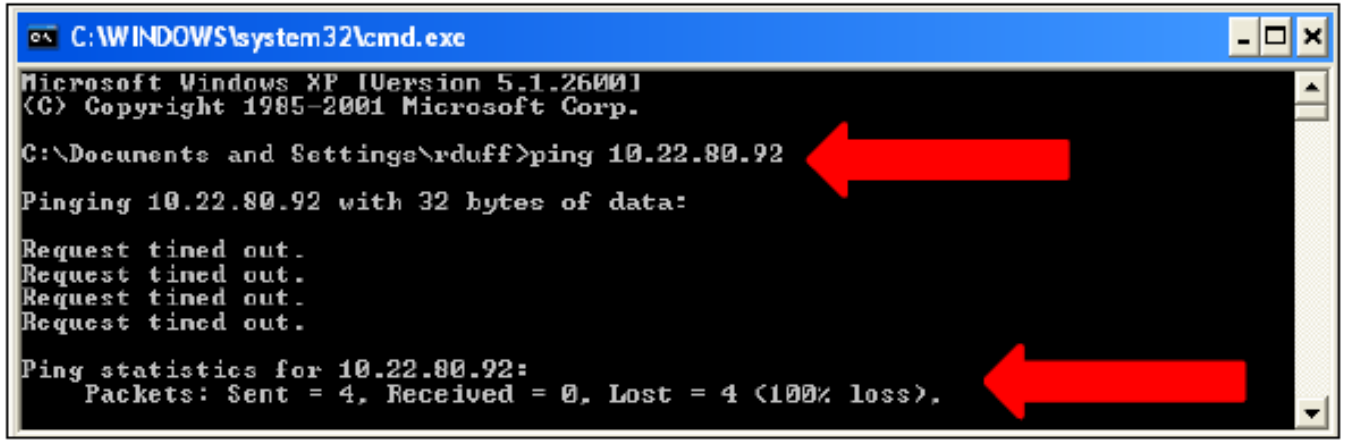
Network Troubleshooting

- **Ping the sensor's IP address** to check whether it is reachable over the network.
- Open a **Command Prompt** and enter the ping command followed by the sensor's IP address.
- **Example:** For a sensor with the IP address 192.168.0.1, enter the following command:

  ping 192.168.0.1



IP Address Detected



IP Address Not Detected

Network Troubleshooting

- Confirm that the **PC's IP address** is correctly configured to communicate with the In-Sight sensor.
- Open a **Command Prompt**, type **ipconfig**, and press Enter to display the PC's IP address and other network configuration details.
- **NOTE:** The PC and sensor should be configured with compatible IP addresses and subnet masks to communicate over the network.

